



Master–slave elimination scheme for arbitrary smooth nonlinear multi-point constraints

Jonas Boungard¹ · Jens Wackerfuß¹

Received: 9 November 2023 / Accepted: 7 February 2024 / Published online: 25 May 2024
© The Author(s) 2024

Abstract

Nonlinear multi-point constraints are essential in modeling various engineering problems, for example in the context of (a) linking individual degrees of freedom of multiple nodes to model nonlinear joints, (b) coupling different element types in finite element analysis, (c) enforcing various types of rigidity in parts of the mesh and (d) considering deformation-dependent Dirichlet boundary conditions. One method for addressing constraints is the master–slave elimination, which offers the benefit of reducing the problem dimension as opposed to Lagrange multipliers and the penalty method. However, the existing master–slave elimination method is limited to linear constraints. In this paper, we introduce a new master–slave elimination method for handling arbitrary smooth nonlinear multi-point constraints in the system of equations of the discretized system. We present a rigorous mathematical derivation of the method. Within this method, new constraints can be easily considered as an item of a “constraint library”; i.e. no case-by-case-programming is required. In addition to the theoretical aspects, we also provide helpful remarks on the efficient implementation. Among others, we show that the new method results in a reduced computational complexity compared to the existing methods. The study also places emphasis on comparing the new approach with existing methods via numerical examples. We have developed innovative benchmarks which encompass all relevant computational properties, and provide analytical and reference solutions. Our findings demonstrate that our new method is as accurate, robust and flexible as the Lagrange multipliers, and more efficient due to the reduction of the total number of degrees of freedom, which is particularly advantageous when a large number of constraints have to be considered.

Keywords Nonlinear multi-point constraints · Master–slave elimination · Penalty method · Lagrange multiplier · Benchmark examples

1 Introduction

Nonlinear multi-point constraints are essential in modeling various engineering problems, for example in the context of (a) linking individual degrees of freedom of multiple nodes to model nonlinear joints, see e.g. [27], (b) coupling different element types in finite element analysis, see e.g. [23], (c) enforcing various types of rigidity in parts of the mesh, see e.g. [5], and (d) considering deformation-dependent Dirichlet boundary conditions. These constraints have to be taken into account in the numerical method selected for the underlying physical problem. In this paper, the finite element method (FEM) is considered. Here, constraints stem from two distinct

sources: On the one hand, there are continuous constraints that act upon the underlying function of the physical problem, e.g. the incompressibility of materials. On the other hand, there are discrete constraints that act point-wise upon the structure, e.g. supports and joints. The constraints can either be taken into account by a special finite element formulation (local level), or by defining additional relationships between the nodal degrees of freedom in addition to the assembled nonlinear system of equations (global level). Additionally, the specific class of constraints that arise due to special material properties can be taken into account in the constitutive relation directly (material point level). Typically, continuous constraints are modeled on the local level whereas discrete constraints are modeled on the global level. However, this is no restriction and in the literature the opposed strategies can be found too: Due to the discretization process in the finite element method every continuous constraint or material point constraint can be reformulated as a discrete constraint, see

✉ Jonas Boungard
boungard@uni-kassel.de

¹ Institute of Structural Analysis, University of Kassel, Kassel, Germany

e.g. [30]. Vice versa, the underlying weak form can be modified to include discrete constraints, see e.g. [27, 44].

In this paper, we focus on the so called nonlinear “multi-point constraints”.¹ These are the constraint functions that describe the nonlinear relationships between the nodal degrees of freedom (discrete constraints acting at the global level). With the arguments from above, this is neither a restriction on discrete nor on continuous constraints. In this paper, we focus on arbitrary smooth nonlinear scleronomous constraints (twice continuously differentiable equality constraints independent of time) and static finite element analysis. This includes both inhomogeneous constraints as well as constraints that additionally depend on a control parameter (e.g. required in the context of actuators).

There are three groups of methods to consider multi-point constraints in finite element analysis:

- A. Penalty method, Lagrange multipliers and their combination in form of the augmented and perturbed Lagrange multipliers
- B. Existing master–slave elimination schemes (also called “transformation method”)
- C. For the sake of completeness, all other, less used methods found in the literature

In the methods of group A, the finite element method with constraints is (from a mathematical point of view) considered as an *optimization problem with constraints*. An excellent overview of the multiplier methods (Lagrange multiplier, augmented Lagrange multiplier, perturbed Lagrange multiplier) and penalty method can be found in the textbook by Belytschko et al. [8]. The governing equations of methods in group A are briefly recalled in Sect. 3. Based on that, their advantages and disadvantages also in comparison to the proposed method will be discussed in Sect. 4.2.

In mathematics, the penalty method is a well established method for optimization problems with constraints. Zienkiewicz [45] introduced the method for constraints in the finite element analysis. His paper is focused on examples of different constraints but do not provide the modified equations for the residual vector and a consistent linearization, which can be found in [8].

In mathematical optimization, Lagrange multipliers are a tool to include equality and inequality constraints. However, they originate from mechanics since they were introduced by Lagrange [29] in the context of dynamics as mechanical quantities. In the “proto-FEM”, Lagrange multipliers were introduced by Irons and Draper [26] for the modeling of supports. This paper can be seen as the predecessor of the

incorporation of multi-point constraints in the finite element analysis with Lagrange multipliers. Here, they were interpreted as mechanical quantities too, as it is often the case in mechanics, e.g. identifying pressure as the Lagrange multiplier in case of incompressibility. Even though Lagrange multipliers are widely used in mechanics, no specific journal paper can be pinned down on introducing Lagrange multipliers for arbitrary smooth nonlinear multi-point constraints. An overview on the necessary equations can be found in the text book [8].

Combining the penalty method and the Lagrange multipliers, Hestens [22] and Powell [33] independently developed the augmented Lagrange multipliers for mathematical optimization. It is widely used in the field of mathematical optimization as well as multi-body dynamics, see for example the review by Wojtyra and Frączek [43]. Later it was introduced to variational problems by Glowinski et al. [20], laying the foundation for using augmented Lagrange multipliers in the finite element analysis. Based on this approach, Bourgat et al. [12] were the first to apply the method to a specific finite element analysis. Although they considered only one specific example, they pointed out that it was intended for illustrative purposes only and that the underlying formalism was generally applicable to constraints. However, they only showed the necessary modification of the underlying function, but did not provide the modification of the residual vector and the consistent linearization. These can be found in [8] as well.

The perturbed Lagrange multipliers are another approach of combining the penalty method and the Lagrange multipliers. The corresponding equations were introduced in a paper by Felippa [18]² as a by-product of developing a method to improve the solution derived by the penalty method by an iterative adaption of the penalty factors. Even though the paper also deals with nonlinear constraints, no consistent linearization is provided. Simo et al. [37] coined the term “perturbed Lagrangian” in a paper on contact mechanics in which the perturbed Lagrangian was developed independently of Felippa’s research. However, this paper was limited to linear constraints. For a generalization to arbitrary nonlinear multi-point constraints, see [8] as well. The equations of perturbed Lagrange multipliers can be reformulated to the penalty method, as shown in “Appendix A”. Nevertheless, in this paper both methods will be considered separately for reasons explained in Sect. 3.

Each of these methods can be either formulated on the finite element level (local level) or at the system level (global level). In the discussion presented in this paper, we focus on the latter. The equations of these existing methods will

¹ Strictly speaking, the term “multi-point constraints” can be misleading because it could imply that the constraint equations act upon nodes (points), rather than dofs.

² Later, in his lecture notes on finite element analysis [19], this method called is “augmented Lagrange multipliers”. This can be misleading because it is different to the above mentioned method.

be recalled in Sect. 3 and discussed in detail in Sect. 4.2. By analyzing the similarities and differences, as well as their advantages and disadvantages with respect to the new approach, we provide new insight for the master–slave elimination for nonlinear constraints.

Before discussing in detail group B (the existing master–slave elimination schemes), group C is presented briefly. Barlow [6] introduced the method of combined solutions which does not modify the stiffness matrix but operates with several right hand sides. The combination of the solution provides the solution of the constrained system. The method is limited to homogenous constraints and requires a partition of the degrees of freedom, that is different from the partition used in master–slave elimination. Xikui and Guoqiang [44] modified the underlying variational problem to consider constraints *a priori*. Webb [42] combined the penalty method and the master–slave elimination to avoid a reduction and a rank deficiency of the resulting matrix. Hueck and Schreyer [24] used orthogonal projection to consider constraints, which requires partitioning of the degrees of freedom in such a way that the resulting vectors are orthogonal and fulfill certain conditions with respect to the constraints. Then a system of linear equations (abbreviated in the following with “linear system”) smaller than the original one is solved for the first part of the dofs. In a second step another linear system is solved for the second part of the dofs. This method requires a scaling parameter to prevent the second linear system from becoming singular. Schreyer and Parsons [35] designed an algorithm for the sequential inclusion of constraints. Saint-Georges et al. [34] proposed a subspace projection method aiming for an efficient solution of the resulting system with iterative solvers. Ho et al. [23] developed a specific method for explicit finite element analysis that is limited to the nonlinear coupling of different finite element types (solid-to-beam, solid-to-shell, shell-to-beam). With the exception of the latter paper (with its strong limitation on coupling in an explicit regime), the above mentioned methods are limited to *linear* multi-point constraints.

Group B, the master–slave elimination, was introduced in the finite element analysis for linear constraints by Green [21]. In the following years, extensive research focusing on its efficient implementation was conducted, especially in the 1970s and 1980s. However, this research was not focused on extending the method to nonlinear constraints. The relevant papers are discussed here briefly. Szabo and Tsai [39] interpreted the finite element analysis with constraints as a minimization problem with constraints and performed a transformation analogous to [21] into an unconstrained minimization problem.³ Curiskis and Valliapan [15] presented

an algorithm for the transformation with less manipulation of the matrix and less values to be stored temporarily, thus providing a more computationally efficient method. Abel and Shephard [1] partitioned the degrees of freedom in slave dofs, master dofs and free dofs that are not involved in any constraint equations aiming for a more efficient method by working with submatrices and a symmetric resulting matrix. Shephard [36] presented an algorithm for performing the necessary transformation during the assembly. Similarly, Chang and Lin [13] performed the transformation on the finite element matrices and vectors instead of the global matrix and vector. Ainsworth [2] developed a transformation method that does not require the separation of slave and master degrees of freedom. As the methods in group C, the previously mentioned methods are limited to *linear* constraints. There exist only few papers on master–slave elimination for nonlinear constraints. The paper by Naraswayamy [31] deals explicitly with *nonlinear* constraints. However, the presented method has two major drawbacks: First, it assumes a certain structure of constraint equations that cannot be obtained for most practical cases. Second, no consistent linearization is provided, thus leading to bad convergence rate in a Newton–Raphson scheme. Similarly, Dhondt [17] presents an elimination scheme, in which the linearized mechanical system without constraints is solved solely for the increment of the master degrees of freedom. Based on this current master dofs, the slave dofs are calculated. This approach lacks proper linearization of the constraint forces, leading to a bad convergence rate. Additionally, the constraints have to be formulated in explicit form for the slave dofs which is too restricting for most applications. Jelenić and Crisfield [27] presented a master–slave elimination technique for joints undergoing large rotations by developing specific finite element types (thus operating at the local level) incorporating such joints. This method is solely limited to the constraint arising from joints and cannot be extended to arbitrary nonlinear multi-point constraints. Areias et al. [5] also presented an algorithm for multi-point constraints but their main focus was the efficient access on sparse systems via a graph-theoretical approach.

From the mathematical point of view, the master–slave elimination can be classified as a “null space method”, also known as a “reduced Hessian method” in optimization, see e.g. [9] for an explanation of these methods in a mathematical context.

Of the described methods, the penalty and multiplier methods as well as the master–slave elimination are used by ANSYS for contact problems [3]. For other constraints, ANSYS is limited to the Lagrange multiplier method with the exception of rigid links, for which the master–slave elimination is available too [4]. However, there are multiple restrictions on the use of the master–slave elimination in this context. In ABAQUS the penalty and multiplier meth-

³ They were the only ones noticing this similarity to the derivation of multiplier and penalty methods but they did not exploit this property any further.

ods are used for contact problems. For other constraints, the master–slave elimination is used as standard, but geometrically nonlinear constraints are incorporated using only their incremental form [16]. This limits the performance of a model with such constraints drastically because quadratic convergence of the Newton–Raphson method is prevented by the inconsistent linearization. Additionally, there are situations like distributed coupling for which Lagrange multipliers must be used instead of the master–slave elimination.

This leaves the research gap of a master–slave elimination scheme capable of handling arbitrary *nonlinear* multi-point constraints with consistent linearization. Additionally, both the rigorous mathematical derivation and a comparison of such a scheme to existing methods are currently missing. It is of high interest to develop such a scheme and compare its performance to penalty and multiplier methods because it has the potential to outperform these methods. It should exhibit the same advantages as the existing linear elimination schemes, namely the exact satisfaction of constraints and the reduction of the problem dimension.

Therefore, the main contributions of this paper are:

- A new numerical method for the consideration of *nonlinear* multi-point constraints in the finite element analysis is presented. It is an extension of the well-established master–slave elimination schemes for *linear* multi-point constraints.
- We present a rigorous mathematical derivation of the new method. For this purpose, the nomenclature was deliberately chosen to be in accordance to the textbook by Belytschko [8]. This makes it possible to compare the different methods on the basis of the resulting equations. This also includes the explicit terms for the constraint forces for all method.
- We present various new benchmarks and offer analytical or reference solutions given the lack of such benchmarks in the literature.
- Based on extensive parameter studies of these benchmarks, we compare the numerical performance of the new method with penalty, Lagrange, augmented Lagrange and perturbed Lagrange methods and analyze the advantages and disadvantages.

The method presented in this paper operates at the global level, not on local level. This has the huge advantage of being independent to the underlying finite element formulation. This allows for an easy implementation of new types of constraints because no modification of existing element types has to be performed, in contrast to the local formulation.

The paper is organized as follows: In Sect. 2, the governing equations and the assumptions are presented. In Sect. 3, the penalty and the multiplier methods are recalled. In Sect. 4,

the new method is derived and compared to existing methods on the basis of the resulting equations. In Sect. 5, different aspects of the implementation are discussed. In Sect. 6, several examples are shown to validate the method and illustrate its benefits. Numerical performance of the new method is compared to existing methods for the handling of nonlinear constraints and the results are discussed. The paper closes with a conclusion and a short outlook in Sect. 7.

2 Governing equations and assumptions

In this section, the governing equations of the problem of nonlinear multi-point constraints in the finite element method are shown. Additionally, the required assumptions are presented and discussed briefly.

Starting point is an arbitrary nonlinear boundary value problem (BVP), e.g. a mechanical problem. It is solved by using the finite element method. In the spatial discretization of the weak form of the BVP, the degrees of freedom $\mathbf{V} \in \mathbb{R}^{n_{\text{dof}}}$ are introduced. To obtain the solution for the dofs $\mathbf{V}$, the minimum of the discretized weak form of the BVP, denoted as $W(\mathbf{V}) \in \mathbb{R}$ has to be found

$$\min_{\mathbf{V}} W(\mathbf{V}) \quad (1)$$

which can be interpreted as an *optimization problem without constraints*. Equation (1) is solved by finding a set of unknowns $\mathbf{V}$ with a gradient equal to zero (derivative test). This leads to the following nonlinear system of equations with the residual vector $\mathbf{R} \in \mathbb{R}^{n_{\text{dof}}}$:

$$\mathbf{R}(\mathbf{V}) := \frac{\partial W(\mathbf{V})}{\partial \mathbf{V}} = \mathbf{0} \quad (2)$$

The nonlinear system Eq. (2) cannot be solved analytically in general. If the Newton–Raphson method is used, Eq. (2) has to be linearized, which leads to the following linear system for solving for the unknown increment of the dofs $\Delta \mathbf{V} \in \mathbb{R}^{n_{\text{dof}}}$

$$\mathbf{K}_T \Delta \mathbf{V} = -\mathbf{R}; \quad \text{with} \quad \mathbf{K}_T(\mathbf{V}) := \frac{\partial \mathbf{R}(\mathbf{V})}{\partial \mathbf{V}} = \frac{\partial^2 W(\mathbf{V})}{\partial \mathbf{V}^2} \quad (3)$$

with the tangential stiffness matrix⁴ $\mathbf{K}_T \in \mathbb{R}^{n_{\text{dof}} \times n_{\text{dof}}}$.

The dofs of the initial BVP are now restricted by a set of n_c constraints $\mathbf{c} = [c_1, \dots, c_i, \dots, c_{n_c}]^T \in \mathbb{R}^{n_c}$ depending nonlinearly on the degrees of freedom:

$$\mathbf{c}(\mathbf{V}) = \mathbf{0} \quad (4)$$

⁴ It should be mentioned that $\mathbf{K}_T$ does not have to be symmetric.

See Table 2 for some examples of nonlinear constraints.⁵ With the nonlinear constraints in Eq. (4), the optimization problem without constraints Eq. (1) is transformed in the following *optimization problem with constraints*:

$$\begin{aligned} \min_{\mathbf{V}} W(\mathbf{V}) \\ \text{subject to } \mathbf{c}(\mathbf{V}) = \mathbf{0} \end{aligned} \quad (5)$$

Equation (5) is also solved using the Newton–Raphson method. For this purpose, we additionally introduce the Jacobian of the constraint $\mathbf{G} \in \mathbb{R}^{n_c \times n_{\text{dof}}}$ and Hessian $\mathbf{H} = [\mathbf{H}_1, \dots, \mathbf{H}_i, \dots, \mathbf{H}_{n_c}]$ with $\mathbf{H}_i \in \mathbb{R}^{n_{\text{dof}} \times n_{\text{dof}}}$:

$$\mathbf{G} := \frac{\partial \mathbf{c}}{\partial \mathbf{V}}; \quad \mathbf{H}_i := \frac{\partial^2 c_i}{\partial \mathbf{V}^2} \quad (6)$$

In addition, we introduce the following assumptions on the constraint equations Eq. (4):

- (A1) There exists a solution $\mathbf{V}$ that satisfies every constraint condition $c_i = 0$.
Mathematically, this can be formulated as $\mathbf{0} = \text{image}(\mathbf{c}(\mathbf{V}))$. This is equivalent to the assumption that the constraints do not contradict each other. The detection of such contradictory constraints is beyond the scope of this paper and will be analyzed in a follow-up paper [11].
- (A2) Every constraint function c_i is twice continuously differentiable.
This assumption ensures that the Jacobian and the Hessian of the constraints exists. This is essential for a consistent linearization (Newton–Raphson scheme).
- (A3) The Jacobian $\mathbf{G}$ has full rank, i.e. $\text{rank}(\mathbf{G}) = n_c$.
This assumption ensures that the constraints are non-redundant. This is a necessary condition for the solvability of the resulting linearized system. For redundant constraints, an additional pre-processing or co-processing for the identification and elimination of such redundancy has to be employed, which will be presented in a follow-up paper [11]. It has to be mentioned that this property is not a necessary condition for the penalty method and the perturbed Lagrange multipliers.

Such assumptions are neglected in the literature on nonlinear multi-point constraints. To the best knowledge of the authors, only Ainsworth [2] formulated such assumptions on the constraints, although his assumptions were limited to linear constraints. However, the assumptions are essential to

ensure that the problem with constraints is solvable. In contrast to [2], no property on the uniqueness of the solution is introduced, as this cannot be ensured for nonlinear finite element analysis with nonlinear constraints.

3 Existing methods

In order to compare the proposed new method with the existing methods, see Sect. 4.2, the penalty method and the multiplier methods are briefly recalled in this section. This also includes the explicit formulas for the calculation of the constraint forces in every method. For the mathematical derivation, the nomenclature was deliberately chosen to be in accordance to the textbook by [8] to provide an ideal basis for comparing the methods.

Each method is based on the same central idea: the transformation of the optimization problem with constraints Eq. (5) in a *modified* optimization problem without constraints, which is solved by finding a set of unknowns $\mathbf{V}$ with a gradient equal to zero (derivative test). Therefore, for every method, its concept for the transformation, the modification of the underlying function W_{method} , the modified residual vector $\mathbf{R}_{\text{method}}$ and the modified linearized system $\mathbf{K}_{\text{method}} \Delta \mathbf{V}_{\text{method}} = -\mathbf{R}_{\text{method}}$ with the modified tangential stiffness matrix $\mathbf{K}_{\text{method}}$ is presented (with “method” as a placeholder for the methods shown below).

3.1 Penalty method (PM)

In the penalty method, the constraints are satisfied approximately by modifying the function W by adding a penalty term

$$W_{\text{PM}}(\mathbf{V}, \beta) = W(\mathbf{V}) + \frac{1}{2} \beta \mathbf{c}(\mathbf{V})^T \mathbf{c}(\mathbf{V}) \quad (7)$$

This term is the weighted square error of the constraints. The weight $\beta \in \mathbb{R}^+$ is called “penalty factor”. Each constraint can either be weighted separately resulting in a weighting matrix with different penalty factors β_i or weighted identically with a single penalty factor β . In the following, without loss of generality, only the latter approach is shown for the sake of clarity.

Derivation of Eq. (7) with respect to the dofs $\mathbf{V}$ results in the modified residual vector $\mathbf{R}_{\text{PM}} \in \mathbb{R}^{n_{\text{dof}}}$:

$$\mathbf{R}_{\text{PM}}(\mathbf{V}, \beta) = \mathbf{R}(\mathbf{V}) + \beta \mathbf{c}(\mathbf{V})^T \mathbf{G}(\mathbf{V}) = \mathbf{0} \quad (8)$$

In Eq. (8), the constraint forces can be identified as $\mathbf{C}_{\text{PM}} := -\beta \mathbf{c}^T \mathbf{G}$.

⁵ An example for linear constraints is the inclined support $\sin(\alpha)u + \cos(\alpha)w = 0$ with constant inclined angle α .

Linearization of Eq. (8) results in the following linear system for the increments of the dofs $\Delta \mathbf{V}$

$$\underbrace{(\mathbf{K}_T + \beta (\mathbf{G}^T \mathbf{G} + c_i \mathbf{H}_i))}_{=: \mathbf{K}_{PM}} \Delta \mathbf{V} = -\mathbf{R}(\mathbf{V}) - \beta \mathbf{c}(\mathbf{V})^T \mathbf{G}(\mathbf{V}) \quad (9)$$

with the resulting stiffness matrix $\mathbf{K}_{PM} \in \mathbb{R}^{n_{dof} \times n_{dof}}$.

3.2 Lagrange multipliers (LM)

It is well known that in the optimum of an optimization problem with constraints, the gradient of the objective function is a linear combination of the gradients of the constraints. The factors of the linear combination are the Lagrange multipliers $\lambda \in \mathbb{R}^{n_c}$. This leads to the following modified function W_{LM} , often called “Lagrange function” or “Lagrangian”:

$$W_{LM}(\mathbf{V}, \lambda) = W(\mathbf{V}) + \lambda^T \mathbf{c}(\mathbf{V}) \quad (10)$$

Derivation of Eq. (10) with respect to the dofs $\mathbf{V}$ results in the modified residual vector $\mathbf{R}_{LM} \in \mathbb{R}^{n_{dof}+n_c}$:

$$\mathbf{R}_{LM}(\mathbf{V}, \lambda) = \begin{bmatrix} \mathbf{R}(\mathbf{V}) + \lambda^T \mathbf{G}(\mathbf{V}) \\ \mathbf{c}(\mathbf{V}) \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (11)$$

In Eq. (11), the constraint forces⁶ can be identified as $\mathbf{C}_{LM} := -\lambda^T \mathbf{G}$.

Linearization of Eq. (11) results in the following linear system for the increments of the dofs $\Delta \mathbf{V}$ and the increments of the Lagrange multipliers $\Delta \lambda \in \mathbb{R}^{n_c}$

$$\underbrace{\begin{bmatrix} \mathbf{K}_T + \lambda_i \mathbf{H}_i & \mathbf{G}^T \\ \mathbf{G} & \mathbf{0} \end{bmatrix}}_{=: \mathbf{K}_{LM}} \begin{bmatrix} \Delta \mathbf{V} \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} -\mathbf{R}(\mathbf{V}) - \lambda^T \mathbf{G}(\mathbf{V}) \\ -\mathbf{c}(\mathbf{V}) \end{bmatrix} \quad (12)$$

with the resulting stiffness matrix $\mathbf{K}_{LM} \in \mathbb{R}^{(n_{dof}+n_c) \times (n_{dof}+n_c)}$.

3.3 Augmented Lagrange multipliers (ALM)

The method of augmented Lagrange multipliers combines the Lagrange multiplier and the penalty method aiming for a relaxation of the problem. Here, both Lagrange multipliers $\lambda \in \mathbb{R}^{n_c}$ and a penalty factor $\beta \in \mathbb{R}^+$ are used to construct the following modified function:

$$W_{ALM}(\mathbf{V}, \lambda, \beta) = W(\mathbf{V}) + \lambda^T \mathbf{c}(\mathbf{V}) + \frac{1}{2} \beta \mathbf{c}^T \mathbf{c} \quad (13)$$

⁶ In the literature, the Lagrange multipliers themselves are sometimes mistakenly referred to as constraint forces.

Derivation of Eq. (13) with respect to the dofs $\mathbf{V}$ results in the modified residual vector $\mathbf{R}_{ALM} \in \mathbb{R}^{n_{dof}+n_c}$ results in:

$$\mathbf{R}_{ALM}(\mathbf{V}, \lambda, \beta) = \begin{bmatrix} \mathbf{R}(\mathbf{V}) + \lambda^T \mathbf{G}(\mathbf{V}) + \beta \mathbf{c}(\mathbf{V})^T \mathbf{G}(\mathbf{V}) \\ \mathbf{c}(\mathbf{V}) \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (14)$$

In Eq. (14), the constraint forces can be identified as $\mathbf{C}_{ALM} := -\lambda^T \mathbf{G} - \beta \mathbf{c}^T \mathbf{G}$.

Linearization of Eq. (14) results in the following linear system for the increments of the dofs $\Delta \mathbf{V}$ and the increments of the Lagrange multipliers $\Delta \lambda$

$$\underbrace{\begin{bmatrix} \mathbf{K}_T + \lambda_i \mathbf{H}_i + \beta (\mathbf{G}^T \mathbf{G} + c_i \mathbf{H}_i) & \mathbf{G}^T \\ \mathbf{G} & \mathbf{0} \end{bmatrix}}_{=: \mathbf{K}_{ALM}} \begin{bmatrix} \Delta \mathbf{V} \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} -\mathbf{R}(\mathbf{V}) - \lambda^T \mathbf{G}(\mathbf{V}) - \beta \mathbf{c}(\mathbf{V})^T \mathbf{G}(\mathbf{V}) \\ -\mathbf{c}(\mathbf{V}) \end{bmatrix} \quad (15)$$

with the resulting stiffness matrix $\mathbf{K}_{ALM} \in \mathbb{R}^{(n_{dof}+n_c) \times (n_{dof}+n_c)}$ and the residual vector⁷.

3.4 Perturbed Lagrange multipliers (PLM)

In contrast to augmented Lagrange multipliers, the method of perturbed Lagrange multipliers is a different combination of the Lagrange multiplier and the penalty method aiming for a regularization of the problem. Here, both Lagrange multipliers $\lambda \in \mathbb{R}^{n_c}$ and a perturbation factor $\epsilon \in \mathbb{R}^+$ are used to construct the following modified function:

$$W_{PLM}(\mathbf{V}, \lambda, \epsilon) = W(\mathbf{V}) + \lambda^T \mathbf{c}(\mathbf{V}) - \frac{1}{2} \epsilon \lambda^T \lambda \quad (16)$$

By derivation of Eq. (16) with respect to the dofs $\mathbf{V}$, the modified residual vector $\mathbf{R}_{PLM} \in \mathbb{R}^{n_{dof}+n_c}$ results in:

$$\mathbf{R}_{PLM}(\mathbf{V}, \lambda, \epsilon) = \begin{bmatrix} \mathbf{R}(\mathbf{V}) + \lambda^T \mathbf{G}(\mathbf{V}) \\ \mathbf{c}(\mathbf{V}) - \epsilon \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} \quad (17)$$

In Eq. (17), the constraint forces can be identified as $\mathbf{C}_{PLM} := -\lambda^T \mathbf{G}$.

Linearization of Eq. (17) results in the following linear system for the increments of the dofs $\Delta \mathbf{V}$ and the increments of the Lagrange multipliers $\Delta \lambda$

$$\underbrace{\begin{bmatrix} \mathbf{K}_T + \lambda_i \mathbf{H}_i & \mathbf{G}^T \\ \mathbf{G} & -\epsilon \mathbf{1} \end{bmatrix}}_{=: \mathbf{K}_{PLM}} \begin{bmatrix} \Delta \mathbf{V} \\ \Delta \lambda \end{bmatrix} = \begin{bmatrix} -\mathbf{R}(\mathbf{V}) - \lambda^T \mathbf{G}(\mathbf{V}) \\ -\mathbf{c}(\mathbf{V}) + \epsilon \lambda \end{bmatrix} \quad (18)$$

with the resulting stiffness matrix $\mathbf{K}_{PLM} \in \mathbb{R}^{(n_{dof}+n_c) \times (n_{dof}+n_c)}$ and the residual vector⁷.

⁷ For the term $-\lambda^T \mathbf{G}(\mathbf{V})$, there is a typographical error in [8] in form of a missing negative sign.

3.5 Remarks on multiplier methods

In the multiplier methods, instead of using the increment of the Lagrange multipliers in the linearized system Eqs. (12), (15) and (18), the multipliers themselves can be used because the right-hand side depends linearly on the multipliers:

$$\begin{bmatrix} \mathbf{K}_T + \lambda_i \mathbf{H}_i & \mathbf{G}^T \\ \mathbf{G} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \Delta \mathbf{V} \\ \lambda \end{bmatrix} = \begin{bmatrix} -\mathbf{R}(\mathbf{V}) \\ -\mathbf{c}(\mathbf{V}) \end{bmatrix} \quad (19)$$

In the implementation, this demands for a separation of degrees of freedom with increments ($\mathbf{V}$ with $\Delta \mathbf{V}$) and without increments (λ).

Based on this, it is possible to transform the perturbed Lagrange multiplier scheme into the penalty method by elimination of the Lagrange multipliers (PLM with elim.), see “Appendix A”. Then, the resulting penalty factor β is equal to the inverse of the perturbation factor ϵ . In this case, the method exhibits the same advantages and disadvantages as the penalty method. Nevertheless, in this paper the perturbed Lagrange multiplier scheme *without elimination* (PLM without elim.) is considered as a separate method because a different behavior between the penalty method and the perturbed Lagrange multipliers without elimination was observed in numerical studies. This is due to the different right-hand sides and the different modified tangential stiffness matrices with different condition number.

4 Master–slave elimination of nonlinear multi-point constraints

4.1 Derivation

In the following, the master–slave elimination scheme for arbitrary smooth nonlinear multi-point constraints is derived. It is shown for the first time, that this derivation is based on the same assumptions and principles as the derivation of the penalty and multiplier methods summarized in Sect. 3.

The central idea of the new approach is based on the calculation of the constraint forces $\mathbf{C} = [C_1, \dots, C_i, \dots, C_{n_{\text{dof}}}] \in \mathbb{R}^{n_{\text{dof}}}$ resulting from the constraints $\mathbf{c}$ introduced in Sect. 2. The constraint forces have to be taken into account in the equilibrium equations Eq. (2). This leads to the modified nonlinear system of equations with the modified residual vector $\mathbf{R}_{\text{mod}}$:

$$\mathbf{R}_{\text{mod}} := \mathbf{R} - \mathbf{C} \quad (20)$$

The negative sign in Eq. (20) is due to a sign convention to be in accordance to the constraint forces identified in Sect. 3 and the nomenclature in [8]. Although the vector of constraint forces $\mathbf{C}$ has the dimension n_{dof} , only the entries belonging to

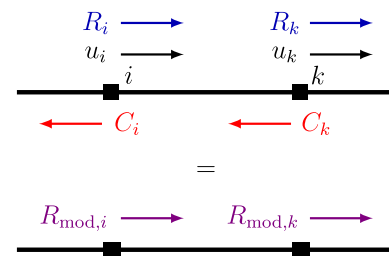


Fig. 1 Illustration of the constraint forces

degrees of freedom involved in constraints can be non-zero. In Fig. 1, this modification of the equilibrium and the residual vector is illustrated. Here, a simple one-dimensional distance constraint between nodes i and k in form of $c(\mathbf{V}_i, \mathbf{V}_k) = u_i - u_k = 0$ is employed.

The residual vector $\mathbf{R}$ is defined as the derivative of the underlying function $W(\mathbf{V})$ with respect to the unknowns $\mathbf{V}$, see Eq. (2). Thus, for the derivation of the modified residual $\mathbf{R}_{\text{mod}}$, the underlying function must be modified. As before, the derivation is based on the same central idea (the transformation of the optimization problem with constraints into the modified optimization problem without constraints) as the methods recalled in Sect. 3.

In the master–slave elimination, the degrees of freedom are partitioned into $n_{\text{dof}} - n_c$ “master dofs”, denoted with m , and n_c “slave dofs”, denoted with s

$$\mathbf{V} = \begin{bmatrix} \mathbf{V}_m \\ \mathbf{V}_s \end{bmatrix} \quad (21)$$

with the subvectors of the master dofs $\mathbf{V}_m \in \mathbb{R}^{n_{\text{dof}} - n_c}$ and the slave dofs $\mathbf{V}_s \in \mathbb{R}^{n_c}$. For each constraint, one slave dof can be identified. For the minimal example in Fig. 1, the slave dof could either be the horizontal displacement u_i at node i or the horizontal displacement u_k at node k . The dofs that are not a slave dof, even the dofs that are not involved in any constraints, are denoted as master dofs.

With the partition in Eqs. (21), (2) and (4) are rewritten and the residual vector is partitioned into the residual vector related to the master dofs $\mathbf{R}_m \in \mathbb{R}^{n_{\text{dof}} - n_c}$ and the residual vector related to the slave dofs $\mathbf{R}_s \in \mathbb{R}^{n_c}$:

$$\begin{aligned} \mathbf{R}_m(\mathbf{V}_m, \mathbf{V}_s) &:= \frac{\partial W}{\partial \mathbf{V}_m} = \mathbf{0} \\ \mathbf{R}_s(\mathbf{V}_m, \mathbf{V}_s) &:= \frac{\partial W}{\partial \mathbf{V}_s} = \mathbf{0} \\ \mathbf{c}(\mathbf{V}_m, \mathbf{V}_s) &= \mathbf{0} \end{aligned} \quad (22)$$

Due to the third assumption on the constraints (A3), see Sect. 2, a selection of slave dofs $\mathbf{V}_s$ can be made that fulfills the following condition concerning the quadratic submatrix $\mathbf{G}_s \in \mathbb{R}^{n_c \times n_c}$ of the Jacobian $\mathbf{G}$:

$$\det(\mathbf{G}_s) \neq 0; \quad \mathbf{G}_s := \frac{\partial \mathbf{c}}{\partial \mathbf{V}_s} \quad (23)$$

The necessity to choose slave degrees of freedom $\mathbf{V}_s$ that fulfill Eq. (23) is a drawback of the master–slave elimination, but this process can be automated. However, this is beyond the scope of this paper and will be analyzed in a follow-up paper [11]. In the following, we assume that an appropriate selection of degrees of freedom has been made.

With the second assumption on the constraints (A2), see Sect. 2, as well as Eqs. (22)₂ and (23), the conditions to apply the *implicit function theorem* are satisfied. Therefore, Eq. (22)₂ can be rewritten in an explicit form for the slave degrees of freedom $\mathbf{V}_s$ in the neighborhood U of the solution

$$\mathbf{c}(\mathbf{V}_m, \mathbf{V}_s) = \mathbf{0} \Leftrightarrow \mathbf{V}_s = \mathbf{c}_{\text{mod}}(\mathbf{V}_m); \quad \mathbf{c}_{\text{mod}} \in C^1(U, \mathbb{R}^{n_c}) \quad (24)$$

with the modified constraint vector $\mathbf{c}_{\text{mod}} \in \mathbb{R}^{n_c}$, depending only on the master dofs $\mathbf{V}_m$. With Eq. (24), the constrained optimization problem Eq. (5) can be transformed into an *unconstrained optimization problem*. This provides the modified function W_{mod} we aimed for to include the constraint forces:

$$\begin{aligned} \min_{\mathbf{V}_m, \mathbf{V}_s} W(\mathbf{V}_m, \mathbf{V}_s) \quad \text{sub. to } \mathbf{c}(\mathbf{V}_m, \mathbf{V}_s) = \mathbf{0} \\ \Leftrightarrow \min_{\mathbf{V}_m} W(\mathbf{V}_m, \mathbf{c}_{\text{mod}}(\mathbf{V}_m)) = \min_{\mathbf{V}_m} W_{\text{mod}}(\mathbf{V}_m) \end{aligned} \quad (25)$$

This unconstrained optimization problem is solved by the derivative test. For differentiation, the chain rule is used resulting in the following relationship

$$\mathbf{0} = \underbrace{\frac{\partial W_{\text{mod}}(\mathbf{V}_m)}{\partial \mathbf{V}_m}}_{=: \mathbf{R}_{\text{mod},m}^T} = \underbrace{\frac{\partial W(\mathbf{V}_m, \mathbf{V}_s)}{\partial \mathbf{V}_m}}_{=: \mathbf{R}_m^T} + \underbrace{\frac{\partial W(\mathbf{V}_m, \mathbf{V}_s)}{\partial \mathbf{V}_s} \frac{\partial \mathbf{V}_s}{\partial \mathbf{V}_m}}_{=: \mathbf{R}_s^T} \quad (26)$$

with the definition of the subvector $\mathbf{R}_{\text{mod},m}$ and the subvectors $\mathbf{R}_m$ and $\mathbf{R}_s$ already defined in Eq. (22). Using the *implicit function theorem* applied in Eq. (24), the derivative of the slave degrees of freedom to the master degrees of freedom $\frac{\partial \mathbf{V}_s}{\partial \mathbf{V}_m}$ can be computed

$$\begin{aligned} \frac{\partial \mathbf{V}_s}{\partial \mathbf{V}_m} &= \frac{\partial \mathbf{c}_{\text{mod}}(\mathbf{V}_m)}{\partial \mathbf{V}_m} = - \left(\frac{\partial \mathbf{c}(\mathbf{V}_m, \mathbf{V}_s)}{\partial \mathbf{V}_s} \right)^{-1} \frac{\partial \mathbf{c}(\mathbf{V}_m, \mathbf{V}_s)}{\partial \mathbf{V}_m} \\ &= -\mathbf{G}_s^{-1} \mathbf{G}_m; \quad \mathbf{G}_m := \frac{\partial \mathbf{c}}{\partial \mathbf{V}_m} \end{aligned} \quad (27)$$

defining the second submatrix of the Jacobian $\mathbf{G}_m \in \mathbb{R}^{n_c \times (n_{\text{dof}} - n_c)}$. Plugging Eq. (27) in Eq. (26) leads to the following nonlinear system of equations:

$$\mathbf{0} = \mathbf{R}_{\text{mod},m} = \mathbf{R}_m - \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{R}_s \quad (28)$$

These $n_{\text{dof}} - n_c$ equations are under-determined for the determination of the n_{dof} degrees of freedom $\mathbf{V}$. A solvable system of equations is obtained by taking into account the n_c constraint equations $\mathbf{c}$ as well. Thus, the resulting nonlinear system of equations describing the initial problem, see Eq. (5), is given by

$$\mathbf{0} = \mathbf{R}_{\text{mod}} := \begin{bmatrix} \mathbf{R}_{\text{mod},m} \\ \mathbf{R}_{\text{mod},s} \end{bmatrix} = \begin{bmatrix} \mathbf{R}_m - \overbrace{\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{R}_s}^{=: \mathbf{C}_m} \\ \mathbf{c} \end{bmatrix} \quad (29)$$

defining the modified residual vector $\mathbf{R}_{\text{mod}} \in \mathbb{R}^{n_{\text{dof}}}$. Using the definition in Eqs. (20) and (29), the constraint forces at the master dofs $\mathbf{C}_m$ can be identified as $\mathbf{C}_m = \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{R}_s$. Similarly, the constraint forces at the slave dofs can be identified with $\mathbf{C}_s = \mathbf{G}_s^T \mathbf{G}_s^{-T} \mathbf{R}_s$. Thus, the constraint forces can be summarized as follows:

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_m \\ \mathbf{C}_s \end{bmatrix} = \begin{bmatrix} \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{R}_s \\ \mathbf{G}_s^T \mathbf{G}_s^{-T} \mathbf{R}_s \end{bmatrix} = \mathbf{G}^T \mathbf{G}_s^{-T} \mathbf{R}_s \quad (30)$$

It has to be mentioned that the constraint forces in Eq. (30) can be computed without additional costs. It is also possible to visualize the constraint forces easily, see Sect. 6 for examples. This disproves certain objections on the difficulties of the master–slave elimination on calculation and visualization of the constraint forces found in the literature, see [6, 23], and in commercial software, see [4].

The nonlinear system Eq. (29) is solved using the Newton–Raphson method. For this purpose, the modified residual vector $\mathbf{R}_{\text{mod}}$ has to be linearized. In this context, the following abbreviations are introduced

$$\mathbf{K}_{\alpha\beta} := \frac{\partial \mathbf{R}_{\alpha}}{\partial \mathbf{V}_{\beta}}; \quad \mathbf{H}_{\alpha\beta,i} := \frac{\partial c_i}{\partial \mathbf{V}_{\alpha} \mathbf{V}_{\beta}} \quad (31)$$

with the indexes $\alpha, \beta = \{m, s\}$. With this, one obtains the following tangential modified stiffness matrix $\mathbf{K}_{\text{mod}} \in \mathbb{R}^{n_{\text{dof}} \times n_{\text{dof}}}$

$$\begin{aligned} \mathbf{K}_{\text{mod}}(\mathbf{V}) &= \begin{bmatrix} \mathbf{K}_{\text{mod},mm} & \mathbf{K}_{\text{mod},ms} \\ \mathbf{K}_{\text{mod},sm} & \mathbf{K}_{\text{mod},ss} \end{bmatrix} \\ &:= \begin{bmatrix} \frac{\partial \mathbf{R}_{\text{mod},m}}{\partial \mathbf{V}_m} & \frac{\partial \mathbf{R}_{\text{mod},m}}{\partial \mathbf{V}_s} \\ \frac{\partial \mathbf{R}_{\text{mod},s}}{\partial \mathbf{V}_m} & \frac{\partial \mathbf{R}_{\text{mod},s}}{\partial \mathbf{V}_s} \end{bmatrix} = \frac{\partial \mathbf{R}_{\text{mod}}(\mathbf{V})}{\partial \mathbf{V}} \end{aligned} \quad (32)$$

with the submatrices

$$\begin{aligned}\mathbf{K}_{\text{mod},m\beta} &= \frac{\partial(\mathbf{R}_m - \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{R}_s)}{\partial \mathbf{V}_\beta} \\ &= \frac{\partial \mathbf{R}_m}{\partial \mathbf{V}_\beta} - \frac{\partial \mathbf{G}_m^T}{\partial \mathbf{V}_\beta} \mathbf{G}_s^{-T} \mathbf{R}_s - \mathbf{G}_m^T \frac{\partial \mathbf{G}_s^{-T}}{\partial \mathbf{V}_\beta} \mathbf{R}_s - \mathbf{G}_m^T \mathbf{G}_s^{-T} \frac{\partial \mathbf{R}_s}{\partial \mathbf{V}_\beta} \\ &= \mathbf{K}_{m\beta} - \sum_{i=1}^{n_c} (\mathbf{H}_{m\beta,i} Y_i) + \mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{s\beta,i} Y_i) \\ &\quad - \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{s\beta} \\ \mathbf{Y} &:= \mathbf{G}_s^{-T} \mathbf{R}_s \\ \mathbf{K}_{\text{mod},s\beta} &= \frac{\partial \mathbf{c}}{\partial \mathbf{V}_m} = \mathbf{G}_\beta\end{aligned}\quad (33)$$

and the helper vector $\mathbf{Y} \in \mathbb{R}^{n_c}$. Inserting Eqs. (23)₂ and (27)₂ in Eq. (32), one obtains the following linear system to calculate the increments of the master and slave degrees of freedom $\Delta \mathbf{V}_m$ and $\Delta \mathbf{V}_s$

$$\begin{bmatrix} \mathbf{K}_{\text{mod},mm} & \mathbf{K}_{\text{mod},ms} \\ \mathbf{G}_m & \mathbf{G}_s \end{bmatrix} \begin{bmatrix} \Delta \mathbf{V}_m \\ \Delta \mathbf{V}_s \end{bmatrix} = \begin{bmatrix} -\mathbf{R}_{\text{mod},m} \\ -\mathbf{c} \end{bmatrix}; \quad \mathbf{K}_{\text{mod}} \Delta \mathbf{V} = -\mathbf{R}_{\text{mod}} \quad (34)$$

which is of the same dimension as the unconstrained problem. The matrix $\mathbf{K}_{\text{mod}}$ is unsymmetric. We refer to the method presented up to this point as “master–slave *modification*”.

Since the master–slave *elimination* aims for a reduction of the problem size, the increment of the slave degrees $\Delta \mathbf{V}_s$ has to be eliminated. This is done by static condensation of Eq. (34). Manipulating Eq. (34)₂, one obtains the following equation for the increment of the slave degrees $\Delta \mathbf{V}_s$ depending on the increment of the master dofs $\Delta \mathbf{V}_m$:

$$\begin{aligned}\mathbf{G}_m \Delta \mathbf{V}_m + \mathbf{G}_s \Delta \mathbf{V}_s &= -\mathbf{c} \\ \mathbf{G}_s \Delta \mathbf{V}_s &= -\mathbf{c} - \mathbf{G}_m \Delta \mathbf{V}_m \\ \Delta \mathbf{V}_s &= -\mathbf{G}_s^{-1} \mathbf{c} - \mathbf{G}_s^{-1} \mathbf{G}_m \Delta \mathbf{V}_m\end{aligned}\quad (35)$$

Inserting Eq. (35) in Eq. (34)₁, one obtains the linearized, reduced system of equations for the master–slave *elimination* with the reduced dimension $n_{\text{dof}} - n_c$

$$\begin{aligned}\mathbf{K}_{\text{mod},mm} \Delta \mathbf{V}_m + \mathbf{K}_{\text{mod},ms} \Delta \mathbf{V}_s &= -\mathbf{R}_{\text{mod},m} \\ \mathbf{K}_{\text{mod},mm} \Delta \mathbf{V}_m + \mathbf{K}_{\text{mod},ms} (-\mathbf{G}_s^{-1} \mathbf{c} - \mathbf{G}_s^{-1} \mathbf{G}_m \Delta \mathbf{V}_m) &= -\mathbf{R}_{\text{mod},m} \\ \underbrace{(\mathbf{K}_{\text{mod},mm} - \mathbf{K}_{\text{mod},ms} \mathbf{G}_s^{-1} \mathbf{G}_m)}_{=: \mathbf{K}_{\text{red}}} \Delta \mathbf{V}_m &= \underbrace{-\mathbf{R}_{\text{mod},m} + \mathbf{K}_{\text{mod},ms} \mathbf{G}_s^{-1} \mathbf{c}}_{=: -\mathbf{R}_{\text{red}}}\end{aligned}\quad (36)$$

with the reduced stiffness matrix $\mathbf{K}_{\text{red}} \in \mathbb{R}^{(n_{\text{dof}}-n_c) \times (n_{\text{dof}}-n_c)}$ and the reduced residual vector $\mathbf{R}_{\text{red}} \in \mathbb{R}^{n_{\text{dof}}-n_c}$. If the tangential stiffness matrix of the unconstrained problem $\mathbf{K}_T$ is symmetric, the resulting reduced matrix $\mathbf{K}_{\text{red}}$ is symmetric too. This is proven in “Appendix B”. For the special case that all constraints depend linearly on the unknowns $\mathbf{V}$, Eq.

(36) coincides with the equations on existing master–slave elimination schemes for linear constraints, see e.g. [36].

All quantities needed for the new master–slave elimination scheme can be summarized as follows:

$$\begin{aligned}\mathbf{K}_{\text{red}} \Delta \mathbf{V}_m &= -\mathbf{R}_{\text{red}} \\ \mathbf{K}_{\text{red}} &= \mathbf{K}_{\text{mod},mm} - \mathbf{K}_{\text{mod},ms} \mathbf{G}_s^{-1} \mathbf{G}_m \\ \mathbf{R}_{\text{red}} &= \mathbf{R}_{\text{mod},m} - \mathbf{K}_{\text{mod},ms} \mathbf{G}_s^{-1} \mathbf{c} \\ \mathbf{K}_{\text{mod},m\beta} &= \mathbf{K}_{m\beta} - \sum_{i=1}^{n_c} (\mathbf{H}_{m\beta,i} Y_i) \\ &\quad + \mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{s\beta,i} Y_i) \\ &\quad - \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{s\beta} \\ \mathbf{Y} &= \mathbf{G}_s^{-T} \mathbf{R}_s \\ \mathbf{R}_{\text{mod},m} &= \mathbf{R}_m - \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{R}_s \\ \mathbf{G}_m &= \frac{\partial \mathbf{c}}{\partial \mathbf{V}_m} \\ \mathbf{G}_s &= \frac{\partial \mathbf{c}}{\partial \mathbf{V}_s}\end{aligned}\quad (37)$$

In summary, these equations in conjunction with the decondensation step in Eq. (35) form the master–slave elimination scheme for nonlinear multi-point constraints.

4.2 Properties and comparison

In the following, the new master–slave elimination scheme is compared to the existing methods presented in Sect. 3. For this, their advantages and disadvantages are briefly recalled.

The penalty method (PM) is easy to implement because the structure of its effective linear system Eq. (9) does not differ from an unconstrained problem Eq. (3). However, it is well known that the method suffers from two major drawbacks related to the choice of the penalty factor: The constraints are never satisfied exactly and small penalty factors violate them heavily, while large penalty factors approximately satisfy the constraints. However, large penalty factors worsen the condition number of the resulting matrix.

It is well known that Lagrange multipliers (LM) satisfy the constraints exactly but the introduction of additional dofs enlarges the dimension of the problem, see Eq. (12). Moreover, the saddle point structure of the resulting system Eq. (12) poses an additional challenge to the equation solver, leading to numerical problems in some relevant cases. Special preconditioners have to be used in such cases [28].

Augmented Lagrange multipliers (ALM) exhibit the same advantages and disadvantages as the Lagrange multipliers. However, they improve the conditioning of the resulting sys-

tem of equations if an appropriate penalty factor is chosen [8].

Since using perturbed Lagrange multiplier with elimination of the multipliers (PLM with elim.) is equivalent to the penalty method, the method exhibits the same advantages and disadvantages. Using perturbed Lagrange multiplier without elimination of the multipliers (PLM without elim.) has not been discussed in the literature. It inherits the advantages of the penalty method and the disadvantages of the penalty method and the Lagrange multipliers. Furthermore, our numerical studies indicate a more robust behavior as an additional advantage, see Sect. 6.4.

Like the existing master–slave elimination schemes for linear constraints, the new master–slave elimination scheme for nonlinear constraint satisfies the constraints exactly and reduces the dimension of the resulting linear system by the number of constraints, see Eq. (36). As mentioned above, the master-elimination requires the selection of an appropriate set of slave dofs. This poses a challenge but can be done with no additional cost [11]. The master–slave elimination changes the structure of the resulting linear system fundamentally. For the implementation in a finite element code, this requires resizing and reallocation of the arrays. Since all matrices involved in the method are sparse, they can be stored efficiently in compressed sparse row format (CSR). Calculations as well as reallocation of these matrices can in turn be carried out very efficiently with external libraries such as MKL (Intel) [25]. Numerical examples show that the change of size and the need for reallocation only have a neglectable impact on the overall computation time.

Special attention has to be paid to the application of constraint methods in situations where an eigenproblem must be solved with the effective matrix. Multiplier methods result *a priori* in an indefinite effective matrix. Thus, this matrix exhibits a number of negative eigenvalues equal to the number of constraints n_c [9].⁸ For different types of eigensolvers, this has the following impact: Type A refers to eigensolvers that calculate all eigenvalues. It has to be mentioned that these eigensolvers are only used for small problems since the computational effort increases significantly for an increasing dimension. Here, the computational effort increases for multiplier methods since the dimension of the effective matrix is larger. Type B refers to eigensolvers that requires lower and upper bounds and an estimation of the number of eigenvalues between these bounds, e.g. the FEAST algorithm [32] available in the MKL [25]. Here, the indefinite structure of the multiplier methods poses an additional challenge for the user to make a good estimation.

⁸ Assumption: The tangential stiffness matrix of the unconstrained problem $\mathbf{K}_T$ is symmetric and positive definite. There are even more negative eigenvalues if $\mathbf{K}_T$ is indefinite.

Type C refers to eigensolvers that compute the n smallest or largest eigenvalues. Typically, n is chosen to be small. For multiplier methods, n must be chosen to be larger than n_c , since there are at least n_c negative eigenvalues, as mentioned above. This leads to an increase in computational time.

In the following, these implications are briefly discussed with respect to two eigenproblems used in mechanics. First, there is the nonlinear stability analysis. Here, the stability criterion is checked for each equilibrium point. The occurrence of negative eigenvalues of a symmetric stiffness matrix indicates instability. This criterion has to be modified for multiplier methods. In this case, a number of negative eigenvalues exceeding the number of constraints n_c indicates instability. For eigensolvers of type B, this requires the correct guess of the lower bound and the number of eigenvalues so that *all* negative eigenvalues will be found. This is cumbersome for the user since their guess might be wrong. Then, the computation has to be restarted. Additionally, this increases computational time since, depending on the number of constraints, a large number of eigenvalues has to be computed. For eigensolvers of type C, the same problem arises with arguments mentioned in the previous paragraph. Second, there is the computation of the instability point and the corresponding eigenform which is the eigenvector related to the eigenvalue equal to zero. For classical stability problems, this eigenvalue equal to zero is the smallest eigenvalue. In contrast, for multiplier methods, there are n_c eigenvalues smaller than the eigenvalue equal to zero. For eigensolvers of type B, this poses no additional challenge since the upper and lower bounds can be chosen to be near zero. For eigensolvers of type C, the indefinite structure requires the computation of more than n_c eigenvalues due to the arguments mentioned in the previous paragraph. Thus, the computational effort increases. Since both the penalty method and the master–slave elimination result in a positive definite effective matrix⁸, they do not pose additional challenges for the eigenproblems.

The properties of the constraint methods are summarized in Table 1.

The comparison shows that the different methods have different dimensions of the resulting linear system. This, combined with additional computational operations, results in different computational complexity of the methods. Section 5.4 provides an in-depth analysis of this.

5 Implementation and computational cost

In this section, several aspects of the implementation of the new master–slave elimination presented in Sect. 4 are illuminated and discussed. This supports the reader to implement the new method correctly and efficiently.

Table 1 Summary of properties of methods

Method	Constraints	$\mathbf{c} = \mathbf{0}$	Dimension	Add. param	matrix ^a
PM	Nonlinear	Approx	n_{dof}	β	Sym., pos. definite
LM	Nonlinear	Exact	$n_{\text{dof}} + n_c$	None	Sym., indefinite
ALM	Nonlinear	Exact	$n_{\text{dof}} + n_c$	β	Sym., indefinite
PLM without elim	Nonlinear	Approx	$n_{\text{dof}} + n_c$	ϵ	Sym., indefinite
Existing MS elim. ^b	Linear	Exact	$n_{\text{dof}} - n_c$	None	Sym., pos. definite
new MS mod. ^b	Nonlinear	Exact	n_{dof}	None	Unsym
new MS elim. ^b	Nonlinear	Exact	$n_{\text{dof}} - n_c$	None	Sym., pos. definite

^a Assumption: The tangential stiffness matrix of the unconstrained problem $\mathbf{K}_T$ is symmetric and positive definite

^b Appropriate selection of slave dofs $\mathbf{V}_s$ necessary. However, this can be automated without additional computational cost [11]

5.1 Incorporating boundary conditions in the context of constraints

Due to the heavy matrix–vector manipulation of the underlying equations, see summary in Eq. (37), the correct incorporation of boundary conditions in the context of constraints is of great importance. Three aspects are to consider: First, the possibility to consider boundary conditions at dofs that are involved in the constraints. Second, the selection of slave and master dofs for dofs affected by boundary conditions. Third, the sequence of incorporating boundary conditions *and* constraints. In the following, these aspects will be discussed separately for Neumann and Dirichlet boundary conditions.

As usual in the finite element method, Neumann boundary conditions are considered in the residual vector $\mathbf{R}$ at the global level. The reduced residual vector $\mathbf{R}_{\text{red}}$ depends on both the subvectors $\mathbf{R}_s$ and $\mathbf{R}_m$, see Eq. (36). For the three aspects to be considered, this has the following implications: First, it is possible to apply Neumann boundary conditions to dofs that are affected by constraints. Second, they do not influence the selection of slave and master dofs. Third, it is merely necessary to include the Neumann boundary conditions in the residual before the constraints are considered.

As usual in the finite element method, Dirichlet boundary conditions are included by manipulation of the residual vector $\mathbf{R}$, the tangential stiffness matrix $\mathbf{K}_T$ and the vector of increments of the dofs $\Delta\mathbf{V}$. For the consideration of the above mentioned three aspects, it is necessary to distinguish between two strategies to consider Dirichlet boundary conditions. In strategy A, they are considered as simple constraints (linear constraints with one dof). In strategy B, they are considered by eliminating the corresponding rows and columns in the assembled linear system (standard procedure in FEM). The strategy A has the following implications on the three aspects to consider: First, it is possible to apply Dirichlet boundary conditions to dofs that are affected by constraints as long as the boundary condition does not contradict the constraints and vice versa. Second, the Dirichlet

boundary conditions influence the choice of slave dofs. Since they fix the value of specific dofs, these dofs are part of the slave dofs automatically. Third, Dirichlet boundary conditions are considered simultaneously to the constraints within the constraint treatment. The strategy B has the following implications on the three aspects to consider: First, it is possible to apply Dirichlet boundary conditions to dofs that are affected by constraints as long as the boundary condition does not contradict the constraints and vice versa. Also, the Dirichlet boundary conditions cannot be applied to slave dofs because these dofs are not available after the master–slave elimination. In practice, this is no restriction if an automatic choice of slave dofs is employed. Then, dofs with a Dirichlet boundary condition will not be chosen to be slave dofs, see [11]. Thus, referring to the second aspect, the Dirichlet boundary conditions influence the choice of slave dofs. Third, the Dirichlet boundary conditions have to be applied *after* the constraints. Otherwise, they might not be satisfied, as has been observed in numerical studies. Although the strategy B might seem irrelevant due to the strategy A, it is of practical importance for existing finite element codes. Here, the incorporation of Dirichlet boundary conditions has already been implemented. An additional implementation of the Dirichlet boundary condition solely for systems with constraints might be cumbersome.

In summary, it can be concluded that degrees of freedom appearing in constraints can also be imposed with Neumann and Dirichlet boundary conditions. Hence, the application of Neumann and Dirichlet boundary conditions does not interfere with the master–slave elimination and the new method is as flexible as the existing methods. This will also be illustrated by a specific example, see Sect. 6.3. The proper sequence for the finite element method with the master–slave elimination is summarized in the following:

1. Setup the quantities for the unconstrained problem, Eq. (3).
2. Consider Neumann boundary conditions (possible at arbitrary dofs).
3. Select master and slave dofs such that the dofs affected Dirichlet boundary conditions are not slave dofs.
4. Calculate the reduced linear system Eq. (36) in order to consider constraints.
5. Consider Dirichlet boundary conditions (possible at master dofs, not possible at slave dofs).
6. Solve reduced linear system Eq. (36).

5.2 Implementation in existing FE-codes

For each constraint, the presented master–slave elimination scheme requires the following three quantities: the constraints $\mathbf{c}$, the constraint Jacobian $\mathbf{G}$ and the constraint Hessian $\mathbf{H}_i$ of all constraints. This has three implications for the implementation in the framework of the finite element code that are discussed in this section.

First, the presented master–slave elimination scheme uses the same quantities for each constraint as the penalty and multiplier methods. This allows the re-use of existing functions for the computation of constraint quantities. No additional functions for the specific constraints have to be implemented in the master–slave elimination. Therefore, the new method can be easily added to existing FE-codes where other constraint methods have already been implemented.

Second, a library concept for the constraints can be employed, similarly to the element and material libraries usually existing in FEM codes. This allows for an easy extension of existing code with respect to new (arbitrary) constraints. In Table 2, the constraints used in this paper are listed with cross-references to the governing equations of the constraint, its Jacobian and its Hessian as well as to the examples which make use of the constraint.

The first and second implication disprove the statement in [19] that the master–slave elimination needs a case-by-case programming for nonlinear constraints.

Third, the calculation of the three constraint quantities can be performed in a similar manner to the global quantities in the finite element method, the residual vector $\mathbf{R}$ and the tangential stiffness matrix $\mathbf{K}_T$. For each constraint, the specific constraint value $c_i \in \mathbb{R}$, the specific constraint Jacobian $\mathbf{G}_i := \frac{\partial c_i}{\partial \mathbf{V}} \in \mathbb{R}^{1 \times n_{\text{dof}}}$ and the specific constraint Hessian $\mathbf{H}_i := \frac{\partial^2 c_i}{\partial \mathbf{V}^2} \in \mathbb{R}^{n_{\text{dof}} \times n_{\text{dof}}}$ can be computed independently of other constraints, similarly to the element residual vector $\mathbf{r}_e$ and element stiffness matrix $\mathbf{k}_e$ in the FEM. Afterwards, these quantities are assembled in the same manner as in the FEM. Here, the assembly is performed over the number of constraints n_c instead of the number of elements n_e . The assembly of the constraint vector $\mathbf{c}$ and the constraint Jaco-

bian $\mathbf{G}$ is even more simple compared to the FEM because no summation has to be performed (no overlapping within each column and row), which is illustrated in the following:

$$\mathbf{c} = \mathbf{A}_{i=1}^{n_c} c_i = [c_1 \dots c_i \dots c_{n_c}]^T$$

$$\mathbf{G} = \mathbf{A}_{i=1}^{n_c} \mathbf{G}_i = [\mathbf{G}_1^T \dots \mathbf{G}_i^T \dots \mathbf{G}_{n_c}^T]^T \quad (38)$$

The constraint Hessian is treated similarly to the stiffness matrix. In all methods, the constraint specific Hessian $\mathbf{H}_i$ is scaled by the constraint specific entry Y_i of a method specific vector $\mathbf{Y} \in \mathbb{R}^{n_c}$ and summed up in the linearized system, see Eqs. (9), (12), (15), (18) and (37). Implementation-wise, this can be computed as an effective Hessian $\hat{\mathbf{H}} \in \mathbb{R}^{n_{\text{dof}} \times n_{\text{dof}}}$ as shown in the following equation:

$$\hat{\mathbf{H}} = \sum_{i=1}^{n_c} \mathbf{H}_i Y_i$$

$$\text{with } \mathbf{Y} = \begin{cases} \beta \mathbf{c}, & \text{for penalty method,} \\ \boldsymbol{\lambda}, & \text{for Lagrange multipliers,} \\ \beta \mathbf{c} + \boldsymbol{\lambda}, & \text{for augmented Lagrange multipliers,} \\ \boldsymbol{\lambda}, & \text{for perturbed Lagrange multipliers,} \\ \mathbf{G}_s^{-T} \mathbf{R}_s, & \text{for master-slave elimination.} \end{cases} \quad (39)$$

For the assembly of all three quantities, exiting functions of an FEM code can be used, simplifying the implementation.

5.3 Coupling and clustering of constraints

Since the constraint Jacobian related to the slave dofs $\mathbf{G}_s$ has to be inverted in the master–slave elimination multiple times, see summary in Eq. (37), its structure has a huge impact on the computational cost. Therefore, it is computationally beneficial to exploit certain structures of $\mathbf{G}$. This structure depends on the type of coupling of the dofs by the constraints. In the following, the structure of zero and non-zero entries (marked by ‘ $\times$ ’) will be depicted. Columns of $\mathbf{G}$ that build the submatrix $\mathbf{G}_s$ are marked in gray. Three types are identified:

Type 1 Fully decoupled constraints. Any dofs involved in one constraint are not involved in any other constraint, resulting in the following exemplary Jacobian:

$$\mathbf{G} = \begin{bmatrix} 0 & \times & \times & 0 & 0 & 0 & 0 & \times & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \times & \times \\ 0 & 0 & 0 & 0 & \times & \times & \times & 0 & 0 & 0 \end{bmatrix};$$

$$\mathbf{G}_s = \begin{bmatrix} \times & 0 & 0 \\ 0 & 0 & \times \\ 0 & \times & 0 \end{bmatrix} \quad (40)$$

Table 2 Nonlinear constraints used in this paper and corresponding cross-references

Constraint	Constraint quantities	examples
Inextensibility	Appendix D.1	Sects. 6.1, 6.4 and 6.5
Force release (2d)	Appendix D.2	Sects. 6.2 and 6.3
Rigid beam (2d)	Appendix D.3	Sect. 6.3
Angular constraint	Appendix D.4	Sect. 6.4

Here, n_c columns with one non-zero entry can be identified, building the Jacobian related to the slave dofs $\mathbf{G}_s$. This matrix can be inverted very easily. For this reason, the master–slave elimination is computationally highly efficient for this type of constraint coupling. For example, a constraint type that falls into this class is an isolated rigid link between two nodes, see Fig. 2, but generally speaking every isolated constraint falls into this class. An example of such a constraint coupling is used in Sect. 6.1.

Type 2 *Partially coupled constraints*. At least one dof involved in each constraint is not involved in any other constraint, resulting in the following exemplary Jacobian:

$$\mathbf{G} = \begin{bmatrix} 0 & \times & \times & 0 & \times & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \times & 0 & 0 & 0 & \times & 0 & 0 & \times \\ 0 & 0 & \times & 0 & \times & \times & \times & 0 & 0 & 0 \end{bmatrix};$$

$$\mathbf{G}_s = \begin{bmatrix} \times & 0 & 0 \\ 0 & 0 & \times \\ 0 & \times & 0 \end{bmatrix} \quad (41)$$

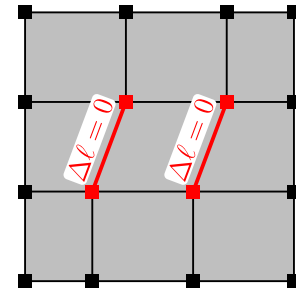
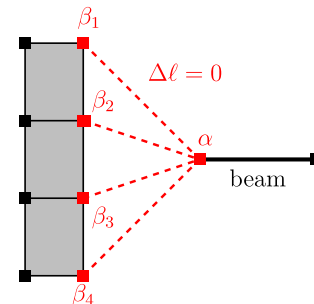
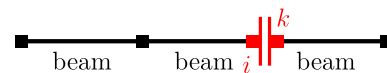
Here, n_c columns with one non-zero entry can be identified, building the Jacobian related to the slave dofs $\mathbf{G}_s$. This matrix can be inverted very easily. For this reason, the master–slave elimination is computationally highly efficient for this type of constraint coupling too. In practice, a rigid coupling of a group of secondary nodes β_i related to one primary node α falls into this class, see Fig. 3.

Type 3 *Fully coupled constraints*. Any dofs involved in one constraint are also involved in at least one other constraint, resulting in the following exemplary Jacobian:

$$\mathbf{G} = \begin{bmatrix} 0 & \times & \times & 0 & \times & \times & 0 & 0 & 0 & 0 \\ 0 & \times & 0 & 0 & 0 & \times & 0 & 0 & 0 & \times \\ 0 & 0 & \times & 0 & \times & \times & 0 & 0 & 0 & \times \end{bmatrix};$$

$$\mathbf{G}_s = \begin{bmatrix} \times & \times & 0 \\ \times & 0 & \times \\ 0 & \times & \times \end{bmatrix} \quad (42)$$

Here, the inversion of the Jacobian related to the slave dofs $\mathbf{G}_s$ is computationally more expensive. However, considering the following two subcases, the computational effort can be reduced: First, there

**Fig. 2** Example for fully decoupled constraints (type 1)**Fig. 3** Example for partially coupled constraints (type 2)**Fig. 4** Example for fully coupled constraints (type 3)

are very small, isolated clusters of fully coupled constraints. For example, a nonlinear joint (like a shear-force release between two nodes i and k , see Sect. 6.2) that is not coupled with other constraints forms such a coupling, see Fig. 4. Here, the matrix $\mathbf{G}_s$ is very small and can be inverted easily. Second, there are larger domains of coupled constraints. Here, building the inverse can be computationally expensive. Since the inverse of the Jacobian related to the slave dofs $\mathbf{G}_s^{-1}$ is used multiple times in one iteration step, a decomposition is the most efficient strategy. Alternatively, iterative solvers capable of handling multiple right-hand sides can be used. An example of such a constraint coupling is given in Sects. 6.2, 6.3 and 6.4.

Furthermore, the constraints are localized in many applications. This allows for the identification of n_{CL} clusters that are built by an arbitrary combination of the constraint coupling types. Figure 5 combines the different examples for coupling types, see Figs. 2, 3 and 4, in one system. Here, three clusters can be identified.

The constraints of each individual cluster are fully decoupled to the constraints of another cluster, resulting in the following block structure of the Jacobian

$$\mathbf{G} = \begin{bmatrix} \mathbf{G}_1 \\ \vdots \\ \mathbf{G}_i \\ \vdots \\ \mathbf{G}_{CL} \end{bmatrix} = [\mathbf{G}_s \mid \mathbf{G}_m] = \begin{bmatrix} \mathbf{G}_{s,1} & \mathbf{0} & \mathbf{0} & \mathbf{G}_{m,1} \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \mathbf{G}_{s,CL} & \mathbf{G}_{m,CL} \end{bmatrix} \quad (43)$$

with the block Jacobian $\mathbf{G}_i$, the block Jacobian related to the slave dofs $\mathbf{G}_{s,i}$ and the block Jacobian related to the master dofs $\mathbf{G}_{m,i}$. The clustering and the resulting block structure are computationally beneficial for three reasons: First, this illustrates the sparsity of the Jacobian $\mathbf{G}$. It can be stored efficiently for example in compressed sparse row format (CSR). Second, the computation of each inverse $\mathbf{G}_{s,i}^{-1}$ can be parallelized. Third, the different types of couplings for each cluster can be used for the efficient computation of the inverse $\mathbf{G}_{s,i}^{-1}$ as presented above.

The identification of the coupling types and clusters can be done by analyzing the CSR structure of the Jacobian. Additional arguments for efficiency by using the sparse structure of the constraint quantities can be found in [5].

5.4 Estimation of computational costs

Compared to the methods presented in Sect. 3, the master–slave elimination relies on heavy manipulation of the residual vector and the tangential stiffness matrix. In order to assess the overall computational costs, the reduction of the dimension also has to be taken into account. In the following, the overall computational cost is estimated. This is done by a computation of the computational complexity (denoted with the Landau symbol $\mathcal{O}$). Here, three computational procedures are considered: First, the calculation of the inverse of the Jacobian related to the slave dofs $\mathbf{G}_s^{-1}$ which is used many times in the master–slave elimination scheme. Second, the calculation of modified and reduced quantities, see summary Eq. (37). Third, the solution of the reduced linear system Eq. (36). For the exact computation of the computational complexity, the use of direct solvers is assumed. The complexity of the three computational procedures is briefly discussed in the following:

1. Considering the worst case scenario, i.e. only fully coupled constraints (type 3, see in Sect. 5.3), then the inversion $\mathbf{G}_s$ is of order $\mathcal{O}((n_c)^3)$, i.e. computationally expensive. However, in many practical cases, there is a mixture of the three types, reducing the actual computational costs. Additionally, for systems with very few constraints, the calculation of the inverse $\mathbf{G}_s^{-1}$ is computationally inexpensive.
2. Only the matrix–matrix multiplications are taken into account because other operations are of neglectable costs.⁹ Here, the following products are of interest:

Product	Computational complexity $\mathcal{O}$
$\mathbf{G}_s^{-1} \mathbf{G}_m$	$(n_c)^2(n_{dof} - n_c)$
$\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{ss}$	$(n_c)^3$
$\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{sm}$	$(n_c)^2(n_{dof} - n_c)$
$\mathbf{K}_{mod,ms} \mathbf{G}_s^{-1} \mathbf{G}_m$	$(n_c)^2(n_{dof} - n_c)$

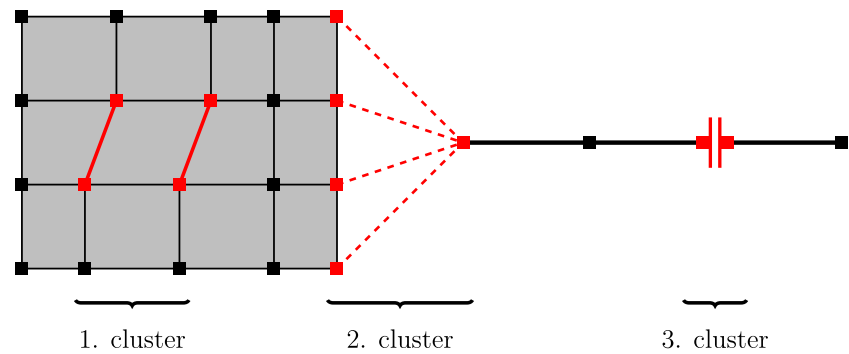
Thus, the total complexity of all matrix–matrix multiplications in the master–slave elimination is $\mathcal{O}(\max((n_c)^2(n_{dof} - n_c), (n_c)^3))$. In the penalty method and for augmented Lagrange multipliers, the only product to consider is $\mathbf{G}^T \mathbf{G}$ which is of complexity $\mathcal{O}((n_{dof})^2 n_c)$. The actual computational costs are lower in practice due to sparsity of the matrices involved. In addition, there exist efficient, parallelized implementations in libraries like the Math Kernel Library (MKL) by Intel [25].

3. The computational cost of the solution of the effective linear system, see e.g. Eq. (36), depends primarily on its dimension. In the master–slave elimination, the dimension is $(n_{dof} - n_c)$. Therefore, the computational complexity is $\mathcal{O}((n_{dof} - n_c)^3)$.

For primarily mathematical reasons for the exact computation of the total computational complexity, two cases for the number of constraints n_c have to be distinguished: Case A, $n_c < n_{dof}/2$, and case B, $n_c > n_{dof}/2$. Note, that the number of constraints n_c cannot be larger than the number of dofs n_{dof} . For both cases, the overall computational complexity of the master–slave elimination is determined. The computational complexity of the three aspects as well as the overall computational complexity are summarized in Table 3. In addition, the computational complexity was computed and is shown for the other methods.

The master–slave elimination exhibits the lowest computational complexity. This is true for an arbitrary number of constraints n_c . Both the penalty method as well as the multiplier methods are computationally more complex. For case B,

⁹ For the ease of comparison, the worst case of a naive implementation of a matrix–matrix multiplication is assumed, neglecting the Strassen algorithm [38] for square matrix multiplication.

Fig. 5 Example for many constraint clusters**Table 3** Overall computational complexity $\mathcal{O}$ of constraint methods for direct solvers

Method	Matrix–matrix mult	$\mathbf{G}_s^{-1}$	Solving linear system	Overall
PM	$(n_{\text{dof}})^2 n_c$	–	$(n_{\text{dof}})^3$	$(n_{\text{dof}})^3$
LM	–	–	$(n_{\text{dof}} + n_c)^3$	$(n_{\text{dof}} + n_c)^3$
PLM	–	–	$(n_{\text{dof}} + n_c)^3$	$(n_{\text{dof}} + n_c)^3$
ALM	$(n_{\text{dof}})^2 n_c$	–	$(n_{\text{dof}} + n_c)^3$	$(n_{\text{dof}} + n_c)^3$
MS elim., case A	$(n_c)^2 (n_{\text{dof}} - n_c)$	$(n_c)^3$	$(n_{\text{dof}} - n_c)^3$	$(n_{\text{dof}} - n_c)^3$
MS elim., case B	$(n_c)^3$	$(n_c)^3$	$(n_{\text{dof}} - n_c)^3$	$(n_c)^3$

the reduction of the computational complexity in the master–slave elimination compared to other methods is even larger compared to case A. The computational complexity gives a good estimate of the actual computational cost. It has to be mentioned that the effective computational cost depends heavily on the specific implementation. For a more practical estimation of the computational costs, two modifications of case A and case B are introduced: Case A*, $n_c \ll n_{\text{dof}}$, and case B*, $n_c = \mathcal{O}(n_{\text{dof}})$. For case A*, as mentioned above, the calculation of the inverse $\mathbf{G}_s^{-1}$ is computationally inexpensive which renders the master–slave elimination to be very efficient. For case B*, the master–slave elimination is also more efficient than existing methods since the same arguments hold as for case B. In summary, the master–slave elimination can be estimated to be more efficient than the other methods, especially for systems with a larger number of constraints.

6 Numerical examples

In this section, several numerical examples are introduced to verify the proposed master–slave elimination and to compare its performance to existing methods capable of handling nonlinear multi-point constraints listed in Table 4. Different types of nonlinear constraints are used in various situations, see Table 2 for an overview. Both fully decoupled (type 1) and fully coupled (type 3) constraints are analyzed in various situations. With these examples, the following properties of the new method are evaluated: (a) accuracy, (b) conver-

gence behavior, (c) robustness, (d) efficiency, (e) impact on the condition number of the resulting matrix.

Since there are no known benchmark tests in the literature that verify these properties in such a fundamental way, the examples presented here can serve as benchmark tests in the future. These examples are deliberately designed to examine all the properties listed above which are analyzed in great detail. Each of the proposed numerical examples is computed in turn using all methods listed in Table 4. Individual symbols are introduced to denote the different numerical methods in all of the following figures, if not stated otherwise. They are listed in the last column of Table 4. All methods presented in Table 4 have been implemented in the in-house research code dockSIM.

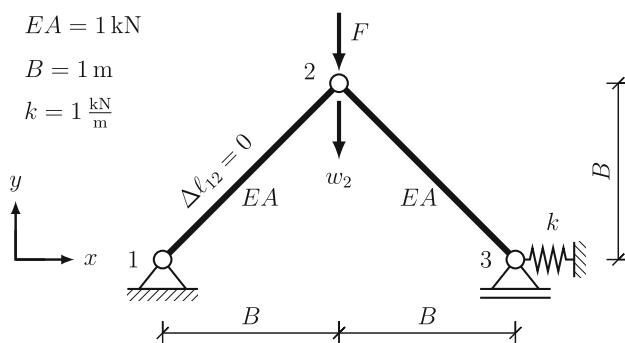
For the sake of brevity, the modified residual vector of all methods are listed in Table 4, see remarks at the start of Sect. 3 and Eq. (20) in Sect. 4.1, is referred to with $\mathbf{R}$ in this section. The unmodified residual vector, see Eq. (2), is only referenced in Sect. 6.3 and is referred to with $\mathbf{R}^*$.

6.1 Snap-through problem with rigid truss

The aim of the first example is to evaluate the proposed master–slave elimination scheme with respect to accuracy and robustness. Additionally, we verify the linearization, see Eq. (36), required in the context of the Newton–Raphson scheme. For this purpose, we analyze a simple snap-through problem build of two linear-elastic trusses (Young’s modulus E , cross-section area A) and a spring (stiffness k), see Fig. 6. Since large strains occur in the truss members during the deformation process, we use the Hencky strain measure with

Table 4 Methods used for numerical examples

Method	Abbr	Equation	Parameter	Symbol and color
Penalty method	PM	Eq. (9)	$\beta = 10$	
penalty method	PM	Eq. (9)	$\beta = 10^3$	
penalty method	PM	Eq. (9)	$\beta = 10^6$	
Lagrange multiplier	LM	Eq. (12)	–	
perturbed Lagrange multiplier	PLM	Eq. (18)	$\epsilon = 10^{-6}$	
augmented Lagrange multiplier	ALM	Eq. (15)	$\beta = 10$	
master-slave elimination	MS elim	Eq. (36)	–	

**Fig. 6** Snap-through problem: geometry and parameters

$\varepsilon = \ln(\ell/L)$ and the work conjugated stress $\sigma = E(\ell/L)\varepsilon$. Additionally, the inextensibility constraint

$$c = \Delta\ell_{12} = L_{12} - \ell_{12} = \|\mathbf{X}_1 - \mathbf{X}_2\| - \|\mathbf{x}_1 - \mathbf{x}_2\| = 0 \quad (44)$$

is introduced for truss 12 to prevent a change in length of the left truss, with the initial length L and the position vector $\mathbf{X}$ in reference configuration and length ℓ and position vector $\mathbf{x}$ in current configuration. The associated Jacobian and Hessian of the constraint Eq. (44) were derived and the results are stated in “Appendix D.1”. The system is loaded at point 2 in y -direction with load F . All geometry and stiffness data are stated in Fig. 6.

We propose this example as a fundamental benchmark to test different methods for handling nonlinear multi-point constraints in a nonlinear finite element analysis. Due to its simplicity, we were able to derive an analytical solution which can be found in the “Appendix C.1”.

For the numerical simulation, the system is discretized with two geometrically exact finite truss elements. Because of the snap-through, the load F is applied by means of a displacement control at node 2. The displacement w_2 is increased incrementally in 10 steps if not stated otherwise.

The numerical results are depicted in Figs. 7, 8, 9 and 10. To verify the implementation and the accuracy of the proposed master–slave elimination scheme, the load–displacement curves were computed with both the new method as well as with the other methods listed in Table 4. They are depicted on the left side in Fig. 7. The results for all methods lie qualitatively in good agreement with the analytical solution. For a quantitative evaluation of the simulation results, we also introduce the relative error measure $e_F = \frac{|F_{\text{ana}} - F_{\text{num}}|}{|F_{\text{ana}}|}$. On this basis, the methods are compared in the diagram on the right side in Fig. 7. This diagram shows that the master–slave elimination is as accurate as the augmented Lagrange multiplier and the Lagrange multiplier method. The relative error is in the range of machine precision. In contrast, the other methods perform several orders of magnitude worse. The error for the perturbed Lagrange multiplier and the penalty methods is approximately of the same order as the perturbation factor ϵ respectively the inverse of the penalty factor β .¹⁰

With this example, we also analyze the capability of the proposed master–slave elimination scheme to calculate the constraint forces. Therefore, the constraint-force-displacement curves were computed with both the new method as well as with the other methods listed in Table 4. They are depicted on the left side in Fig. 8. The results for all methods lie qualitatively in good agreement with the analytical solution, with the exception of the penalty method with $\beta = 10$. Here, the deviation of the penalty method with $\beta = 10$ from the analytical solution is obvious compared to the load–displacement diagrams in Fig. 7. For a quantitative evaluation of the simulation results, we also introduce the relative error measure $e_C = \frac{|C_{\text{ana}} - C_{\text{num}}|}{|C_{\text{ana}}|}$. On this basis, the methods are compared in the diagram on the right side

¹⁰ The results for the fifth load step $w_2 = -1$ m are not depicted because the analytical solution is zero and no relative error can be defined here.

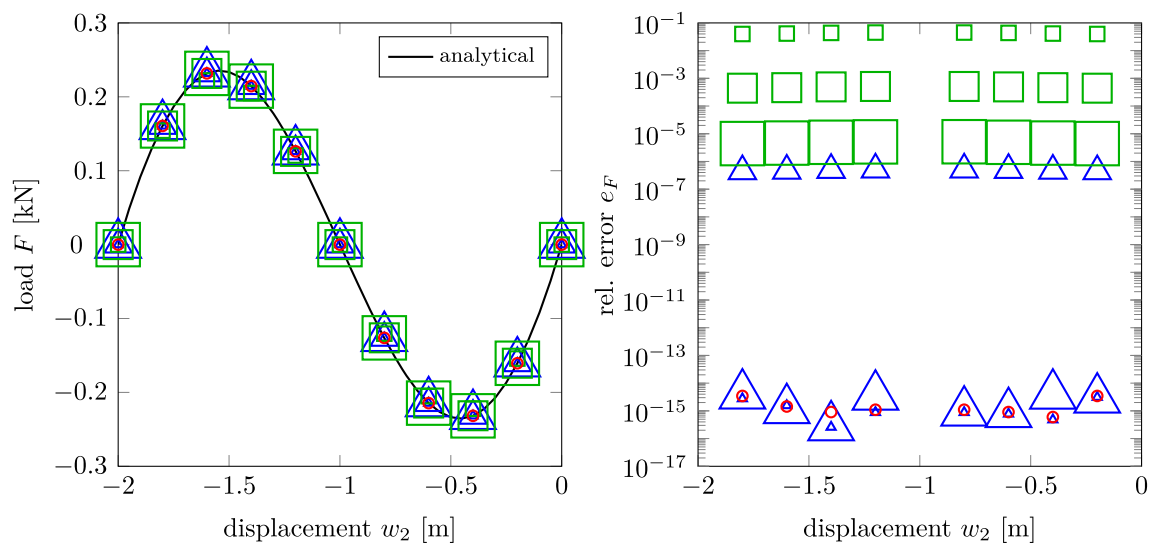


Fig. 7 Snap-through problem: load–displacement diagrams (left) and associated relative error diagram (right); for legend, see Table 4

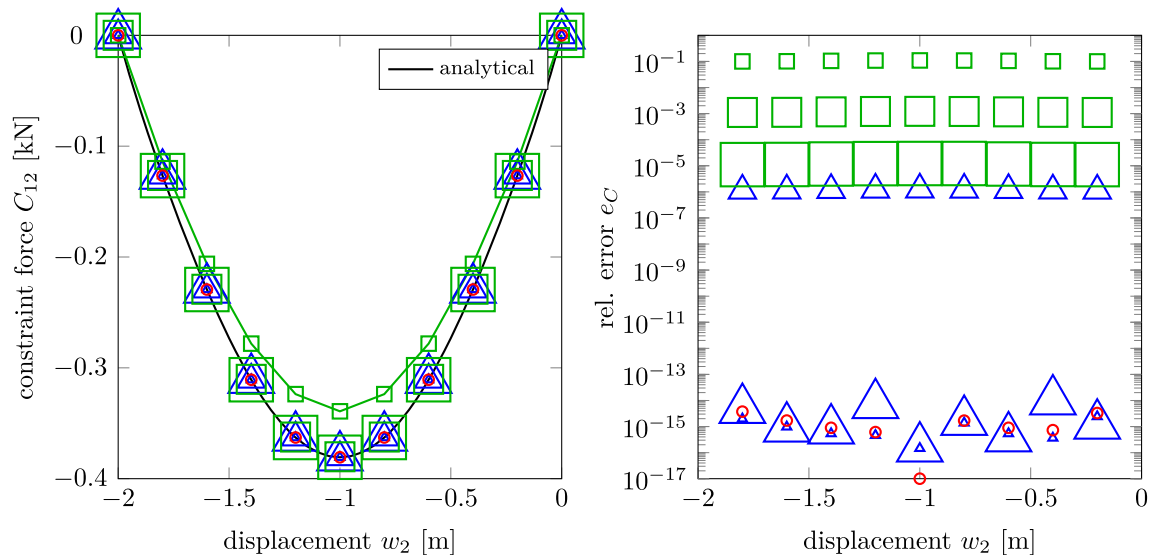


Fig. 8 Snap-through problem: constraint–force–displacement diagram (left) and associated relative error diagram (right); for legend, see Table 4

in Fig. 7. This diagram shows that the master–slave elimination is as accurate as the augmented Lagrange multiplier and the Lagrange multiplier method. The relative error is in the range of machine precision. In contrast, the other methods perform several orders of magnitude worse. The error for the perturbed Lagrange multiplier and the penalty methods is approximately of the same order as the perturbation factor ϵ respectively the inverse of the penalty factor β .

To verify the robustness of the proposed master–slave elimination scheme, additional parameter studies are performed for different load step sizes. The resulting load–displacement diagrams are depicted in Fig. 9 and show that only 5 load steps are required for the simulation of the entire snap-through process. It has to be mentioned that this prop-

erty will also be illustrated in a more complex example with more constraints, see Sect. 6.3.

To analyze the convergence behavior of the proposed master–slave elimination scheme within the Newton–Raphson scheme, the Euclidean norm of the residual vector $\|\mathbf{R}\|$ within the iterative process is computed with both the new method as well as with the other methods listed in Table 4. It is depicted in Fig. 10 (representative for 2 of the 10 load steps). For a comparison, a curve representing the optimal convergence rate of $p = 2$ is also shown. In the numerical simulation, the Newton–Raphson scheme is aborted as soon as the relative criterion $\|\mathbf{R}\|^{(k)} / \|\mathbf{R}\|^{(1)} \leq 10^{-12}$ is satisfied (the superscripts indicating the iteration step). Figure 10 shows that the master–slave elimination as well as

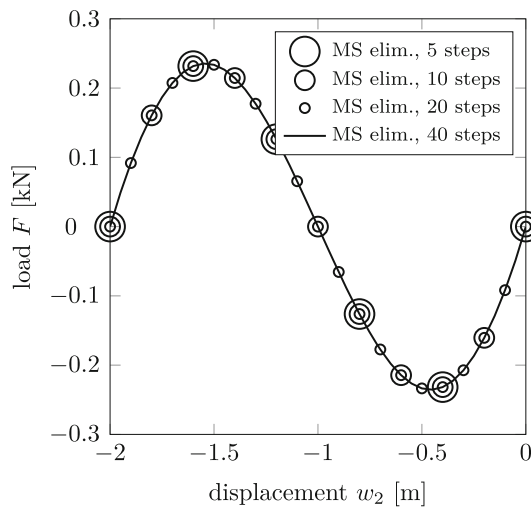


Fig. 9 Snap-through problem: load–displacement diagrams for master–slave elimination with different load steps

the other methods exhibit optimal, quadratic convergence. However, the convergence behavior at the beginning of each load step is different. The multiplier methods and the master–slave elimination exhibit the same behavior while the penalty methods start and end at higher residual norm depending on the penalty factor. This is a key difference between the penalty method and the perturbed Lagrange multiplier variant *without elimination*. Besides the residual norm itself, the number of iteration steps per load step n_{ite} is depicted for certain load steps. Both the master–slave elimination scheme as well as the multiplier methods and the penalty method with $\beta = 10^3$ need $n_{\text{ite}} = 4$ iteration steps. For the load steps that are not depicted in Fig. 10, the same behavior can be observed. The penalty methods with different penalty factors exhibit a similar behavior. For the sake of completeness, it should be mentioned that for $\beta = 10$ one of the previous load steps needs one iteration step less, and for $\beta = 10^6$ this is the case in two previous load steps. Nevertheless, the efficiency and robustness is very similar and the master–slave elimination scheme does not exhibit a different convergence behavior compared to the other methods.

The major results of this numerical example can be summarized and interpreted as follows: The master–slave elimination is accurate and robust. Additionally, one obtains optimal, quadratic convergence if a Newton–Raphson scheme is used to solve the nonlinear system of equations.

6.2 Rolling up a beam with nonlinear shear-force release

The aim of the second example is to evaluate the proposed master–slave elimination scheme with respect to accuracy and robustness in the context of coupling constraints with rotational dofs and strong nonlinearity. For this purpose, we

analyze a planar elastic beam (stiffnesses EA , EI , GA_s), see Fig. 11. The beam is clamped at the left side and is supported by a clamped sleeve on the right side. The beam is split in two parts at node 2 and 3. These nodes are connected by a nonlinear shear-force release which can be described by the following two coupled constraints

$$\begin{aligned} c_1 &= \varphi_2 - \varphi_3 = 0; \\ c_2 &= \mathbf{n}_2(\varphi_2)^T \begin{bmatrix} u_2 \\ w_2 \end{bmatrix} - \mathbf{n}_3(\varphi_3)^T \begin{bmatrix} u_3 \\ w_3 \end{bmatrix} = 0 \end{aligned} \quad (45)$$

with the axial vector $\mathbf{n}$, horizontal and vertical displacements u and w and the rotation φ . The first constraint Eq. (45)₁ depends linearly on the dofs. The second constraint Eq. (45)₂ depends nonlinearly on the dofs. The associated Jacobian and Hessian of the second constraint Eq. (45)₂ were derived and the results are stated in “Appendix D.2”. The constraints are fully coupled (type 3), see Sect. 5.3. The system is loaded at point 2 with moment M . All geometry and stiffness data are stated in Fig. 11. The load is increased until the system exhibits very large displacements and rotations.

We propose this example as a fundamental benchmark to test different methods for handling fully coupled nonlinear multi-point constraints in a nonlinear finite element analysis. Due to its simplicity, we were able to derive an analytical solution which can be found in the “Appendix C.2”. The evaluation of the equations shows that the analytical solution becomes singular for a load of $M = \pi$ kN m and $M = 3\pi$ kN m.

For the numerical simulation, the system is discretized with 8 two-node geometrical exact finite Timoshenko beam elements (unified reduced integration). This coarse discretization was chosen because we are focusing on the constraints rather than the solution for the beam itself, and this discretization is sufficient for representing the rolling-up which can be of course approximated even better with more finite elements. The moment M is applied by means of a rotation control at node 2. The rotation φ_2 is increased incrementally with two different load steps strategies: In strategy A, 29 equidistant load steps are used to avoid a load corresponding to singularity while at the same time the load is increased to close to singularity. The 7th load step corresponds to a load $M = \frac{28}{29} \pi$ kN m which is 97% of the load corresponding to the first singularity. In strategy B, the 10 non-equidistant load steps $\varphi_2 = \{1/4\pi; 23/48\pi; 25/48\pi; 3/4\pi; \pi; 5/4\pi; 71/48\pi; 73/48\pi; 7/4\pi; 2\pi\}$ are used (again to avoid loads corresponding to a singularity) while using a larger load step size to test for robustness. It has to be mentioned that there is a linear relation between the rotation φ_2 and the moment M : $M = 2\varphi_2$. During the complete FE-simulation for all methods, we observe a quadratic rate of convergence in the Newton–Raphson scheme if not stated otherwise.

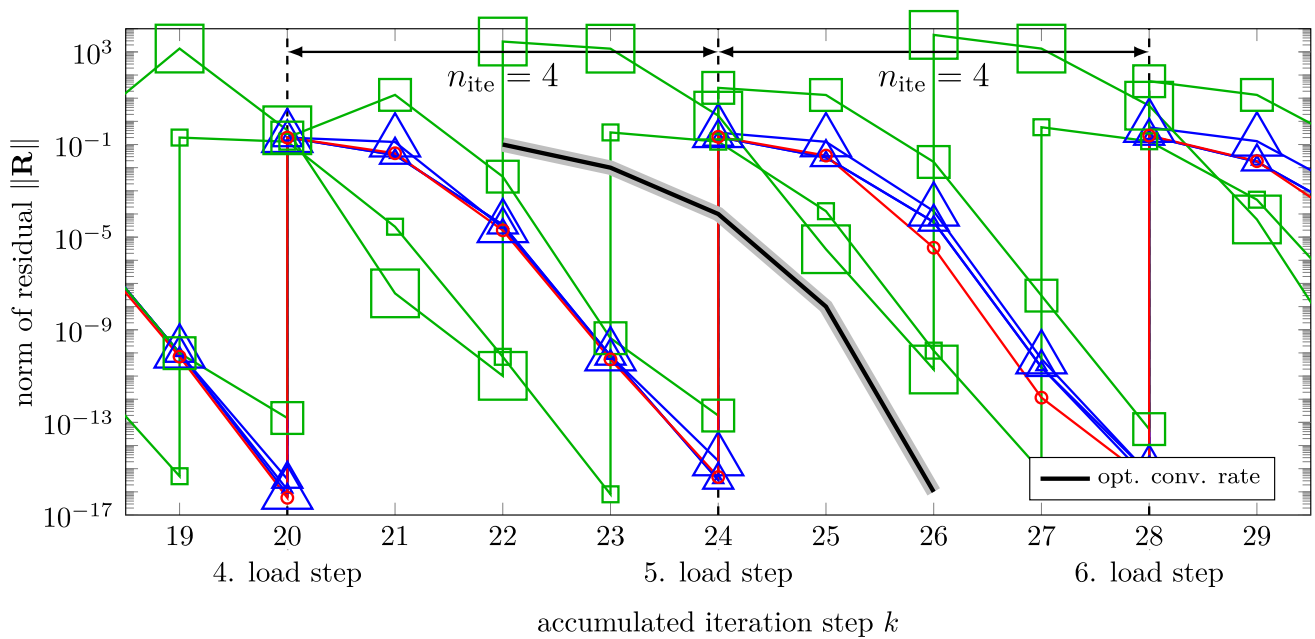
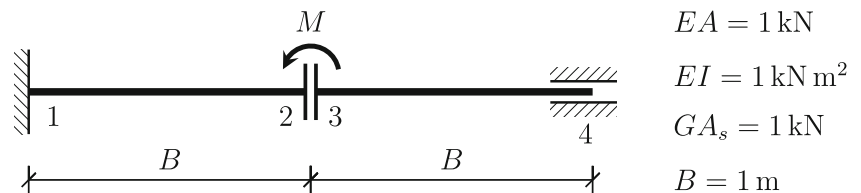


Fig. 10 Snap-through problem: convergence behavior for 2 of 10 load steps; for legend, see Table 4

Fig. 11 Rolling-up of beam: geometry and parameters



The numerical results are depicted in Figs. 12, 13 and 14. To point out the large deformation and to introduce quantities for an error analysis, the analytical deformation as well as the numerical solution computed with the proposed master–slave elimination scheme using load step strategy B are depicted to scale in Fig. 12. Here, the strong nonlinearity of the system can be observed with both large displacements and rotations. The solution for the proposed master–slave elimination scheme lies qualitatively in good agreement with the analytical solution. Only the results for load steps very close to a singularity ($M = \{23/24\pi \text{ kN m}; 25/24\pi \text{ kN m}; 71/24\pi \text{ kN m}; 73/24\pi \text{ kN m}\}$) exhibit differences visible in the figure between the analytical and the numerical solution, see Fig. 12. At the shear-force release, three discontinuities can be identified, that are depicted at the top left of Fig. 12: The rotational discontinuity $\Delta\varphi$, which has to be zero due to Eq. (45)₁, the horizontal discontinuity $\Delta u'$ in the local coordinate system of the shear-force release,¹¹ which has to be 0 due to constraint Eq. (45)₂,

¹¹ Because these values are zero in the analytical solution, only the absolute error can be calculated for $\Delta\varphi$ and $\Delta u'$ for the assessment of numerical solution.

and the vertical discontinuity $\Delta w'$ in the local coordinate system of the shear-force release.

To analyze the behavior close to the singularity, the load–displacement curves were computed with both the new method as well as with the other methods listed in Table 4 for both load step strategies. The results for strategy A are depicted on the left side in Fig. 13. Here, the strong nonlinear behavior of the system can be observed too. For the singularity for a load of $M = \pi \text{ kN m}$ and $M = 3\pi \text{ kN m}$, the displacement u_3 goes to infinity. The results of all methods with the exception of the penalty method with $\beta = 10$ and $\beta = 10^3$ lie qualitatively in good agreement with the analytical solution. The large difference between the configuration before and after a singularity pose no problem for the master–slave elimination as well as for the multiplier methods and the penalty method with $\beta = 10^6$. For a penalty method with $\beta = 10$ and $\beta = 10^3$, the deformation is drastically different to the analytical solution, see left side of Fig. 13. Additionally, for a penalty factor of $\beta = 10^3$ the solution diverges for a load of $M > \frac{24}{29}\pi \text{ kN m}$. The results for strategy B are depicted on the right side in Fig. 13. The results of all methods with the exception of the penalty method with $\beta = 10$ and $\beta = 10^3$ lie qualitatively in good agreement with the analytical solution. Again, the large difference in the defor-

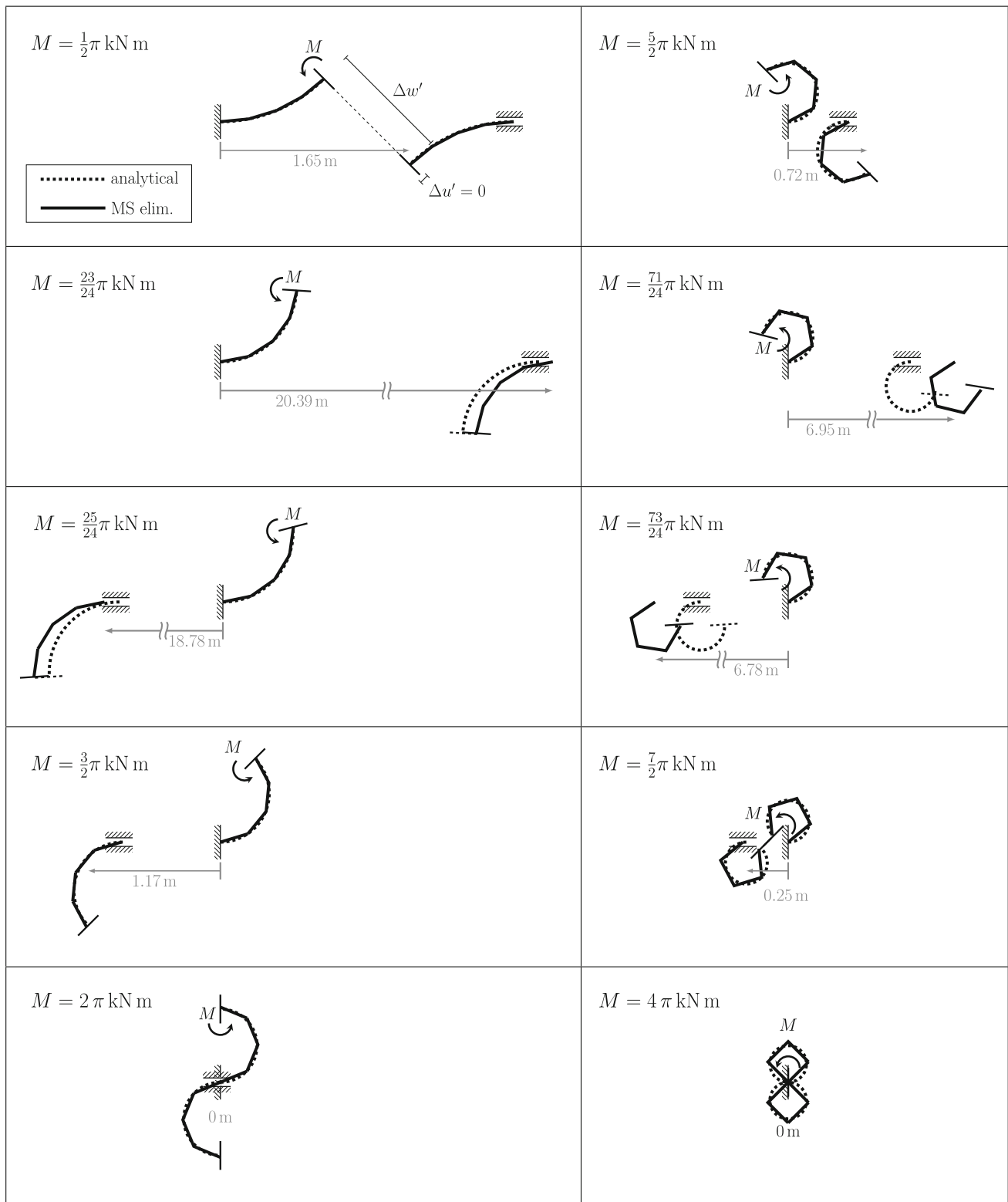


Fig. 12 Rolling-up of beam: true to scale analytical deformation and numerical deformation (MS elim.) with coarse mesh with only 8 finite beam elements (load step strategy B)

mation between the load steps, see Fig. 12, pose no problem for the master–slave elimination as well as for the multiplier methods and the penalty method with $\beta = 10^6$. For a penalty method with $\beta = 10$ and $\beta = 10$, the deformation is drastically different to the analytical solution again, see right side of Fig. 13.

For a quantitative evaluation of the simulation results, the following error measures were computed: The absolute error for the horizontal gap $e_{\Delta u'} = |\Delta u'_{\text{num.}} - \Delta u'_{\text{ana.}}|$, depicted on the top of Fig. 14, the relative error for vertical gap $e_{\Delta w'} = \frac{|\Delta w'_{\text{num.}} - \Delta w'_{\text{ana.}}|}{|\Delta w'_{\text{ana.}}|}$, depicted in the middle of Fig. 14 and the absolute error for rotational gap $e_{\Delta \varphi} = |\Delta \varphi_{\text{num.}} - \Delta \varphi_{\text{ana.}}|$, depicted on the bottom of Fig. 14. For the horizontal gap $\Delta u'$, the error is of similar magnitude for all methods and lies in the range of machine precision. The error for perturbed Lagrange and the penalty method with $\beta = 10$ is slightly worse than for the master–slave elimination, the Lagrange multipliers and the augmented Lagrange multipliers. For the vertical gap $\Delta w'$, the same behavior as for the load–displacement diagrams in Fig. 13 can be observed. The master–slave elimination is as accurate as the Lagrange multiplier and the augmented Lagrange multiplier method and the remaining error can be attributed to the discretization error of the finite element solution. For the rotational gap, the master–slave elimination, the Lagrange multipliers and the augmented Lagrange multipliers satisfy the constraint exactly with $\Delta \varphi = 0$, even taking machine precision into account. The error for the perturbed Lagrange multiplier and the penalty methods is several orders of magnitude larger.

The major results of this numerical example can be summarized and interpreted as follows: The master–slave elimination is accurate and satisfies the constraints exactly, as analytically expected, see Eq. (29) in Sect. 4.1.

6.3 Coupling rigid beam with elastic beam with nonlinear shear-force release

The aim of the third example is to show that it is even possible in the proposed master–slave elimination scheme to additionally apply Neumann boundary conditions or homogenous and inhomogeneous Dirichlet boundary conditions at the dofs involved in constraints. In this context, this example can be seen as a general benchmark. In addition, the accuracy of the proposed master–slave elimination scheme in comparison with other methods, its robustness, its efficiency and the convergence behavior of the constraint norm within the Newton–Raphson scheme are evaluated. For this purpose, we analyze a planar frame with a vertical and a horizontal elastic beam (stiffnesses EA , EI , GA_s), see Fig. 15. The system is supported at the bottom of the vertical beam and the right end of the horizontal beam. The beams are connected by a nonlinear shear-force release, see Eq. (45) and remarks in

Sect. 6.2. The described elastic beam structure is modified subsequently in such a way that the vertical beam becomes rigid due to additional constraints. These constraints will be presented and discussed in the course of this example.

All geometry and stiffness data are stated in Fig. 15. For the load, three cases are analyzed: In load case A, the system is loaded at point 5 with load F in vertical direction. In load cases B and C, the system is loaded at point 3 in horizontal direction. To avoid the bifurcation of the beam structure at the beginning of the deformation, the position of the support on the right side is shifted downwards by $\Delta = 0.02$ m for load case B and C, see Fig. 15. The load is increased until the system exhibits large displacements and rotations.

For the numerical simulation, the vertical beam is discretized with 2 and the horizontal beam is discretized by 10 two-node geometrical exact finite timoshenko beam elements (unified reduced integration). In total, the number of dofs is $n_{\text{dof}} = 50$ and the corresponding number of constraints is $n_c = 8$. For load case A, the load F is applied by means of a load control at node 5. The load is changed over pseudo time τ by the sinusoidal function $F(\tau) = 100 \text{ kN} \cdot \sin(2\pi\tau)$ with pseudo time τ in the range of $[0; 1]$. The pseudo time and thus the load itself are increased incrementally. Different numbers of load steps are used in the course of the analysis and stated in the respective segment. For load case B, a horizontal load F is applied by means of a load control at node 3. The load F is increased incrementally in 80 steps. For load case C, this load is applied by means of a displacement control at node 3. The horizontal displacement u_3 is increased incrementally in 80 steps.

The rigidity of the two vertical beam elements is modeled by the following three coupled constraints per beam element:

$$\begin{aligned} c_1 &= \Delta \ell_{12}(u_1, w_1, u_2, w_2) = 0; \\ c_2 &= \varphi_1 - \varphi_2 = 0; \\ c_3 &= \varphi_1 - \psi_{12}(u_1, w_1, u_2, w_2) = 0; \\ c_4 &= \Delta \ell_{23}(u_1, w_1, u_2, w_2) = 0; \\ c_5 &= \varphi_2 - \varphi_3 = 0; \\ c_6 &= \varphi_2 - \psi_{23}(u_2, w_2, u_3, w_3) = 0 \end{aligned} \quad (46)$$

The first constraint Eq. (46)₁ depends nonlinearly on the horizontal and vertical displacements u and w and prevents the change of length $\Delta \ell_{12}$ between the nodes 1 and 2 respectively the change of length $\Delta \ell_{23}$ between the nodes 2 and 3, see remarks in Sect. 6.1. The combination of the second constraint Eq. (46)₂ and the third constraint Eq. (46)₃ enforces the bending and shearing rigidity. The second constraints Eq. (46)₂ depends linearly on the rotation φ and ensures the same nodal rotation of nodes 1 and 2 respectively nodes 2 and 3. The third constraint Eq. (46)₃ depends nonlinearly on the displacements and rotations, which will be explained and illustrated more in detail in “Appendix D.3”, and ensures

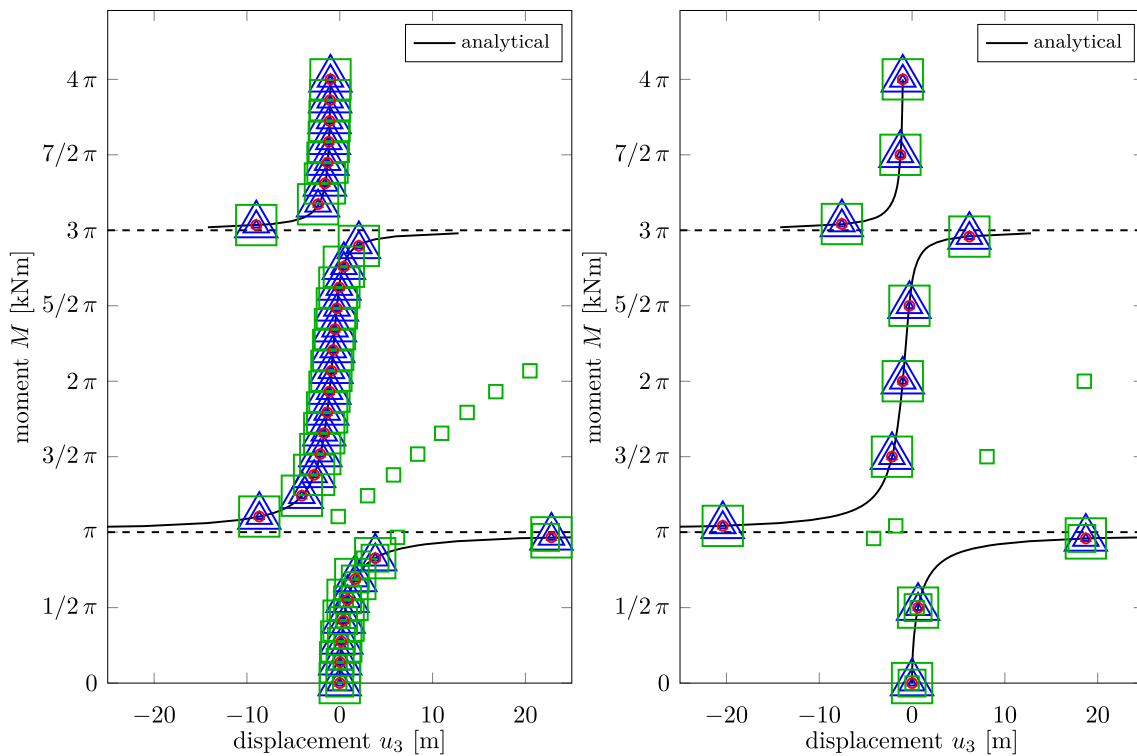


Fig. 13 Rolling-up of beam: load–displacement diagrams for load step strategy A (left) and B (right); for legend, see Table 4

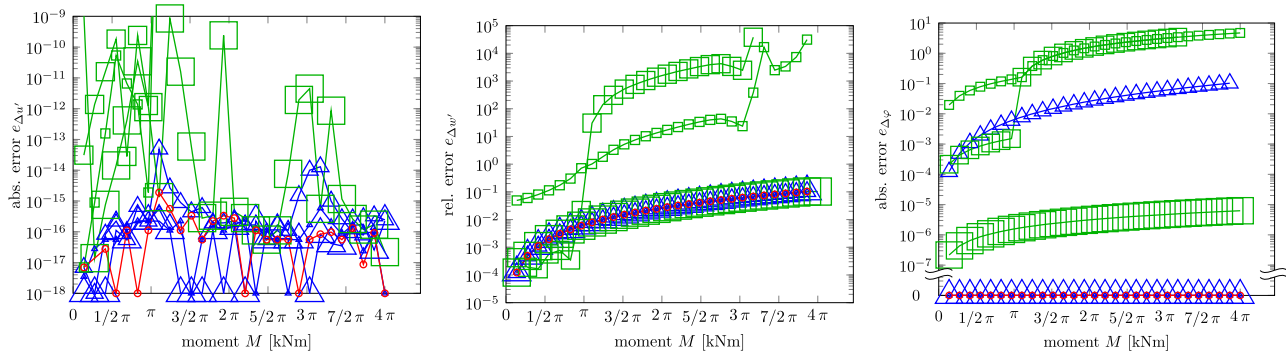


Fig. 14 Rolling-up of beam: absolute error for gap $\Delta u'$ (top), relative error for gap $\Delta w'$ (middle) and absolute error for gap $\Delta \varphi$ (bottom); for legend, see Table 4

the identity of the beam rotation ψ and the nodal rotation φ . The associated Jacobian and Hessian of the third constraint Eq. (46)₃ were derived and the results are stated in “Appendix D.3” too. The constraints associated to the shear-force release and the rigidity of the vertical beam are fully coupled (type 3), see Sect. 5.3. During the complete FE-simulation for all methods, we observe a quadratic rate of convergence in the Newton–Raphson scheme if not stated otherwise.

The numerical results are depicted in Figs. 16, 17, 18, 19 and 20. To show that it is even possible in the proposed master–slave elimination scheme to additionally apply Neumann boundary conditions or homogenous and inhomogeneous Dirichlet boundary conditions at the dofs involved

in constraints, see Sect. 5.1, the load–displacement curves for load cases B and C of the example are computed with both the new method as well as the Lagrange multipliers for reference. The resulting load–displacement diagrams are depicted in Fig. 16. The master–slave elimination yields the same results as the Lagrange multipliers. Both the Neumann boundary condition and the homogeneous Dirichlet boundary condition (support at the bottom) as well as the inhomogeneous Dirichlet boundary condition (prescribed displacement u_3) affecting dofs participating in constraints can be processed correctly in the master–slave elimination.

Additionally, to verify the nonlinear behavior of the system and the accuracy of the proposed master–slave elimination scheme, the load–displacement curves for load case

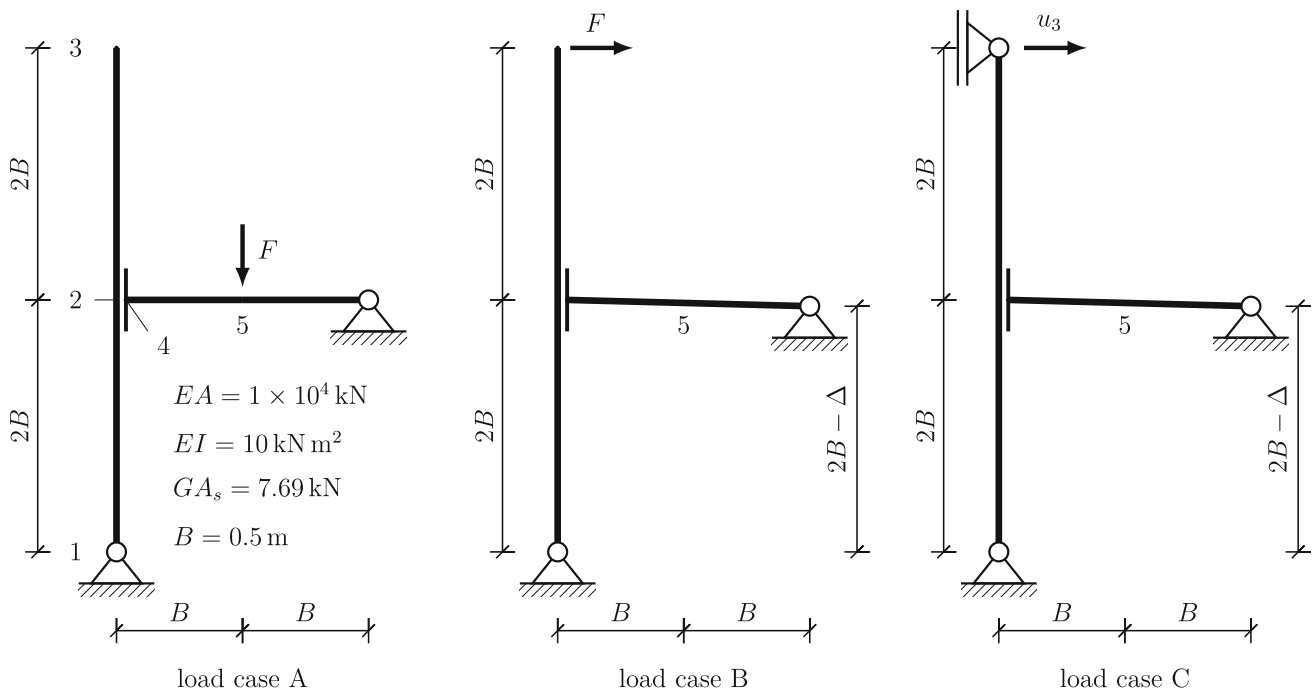


Fig. 15 Coupled rigid and elastic beam: geometry and parameters. Three different load cases are investigated

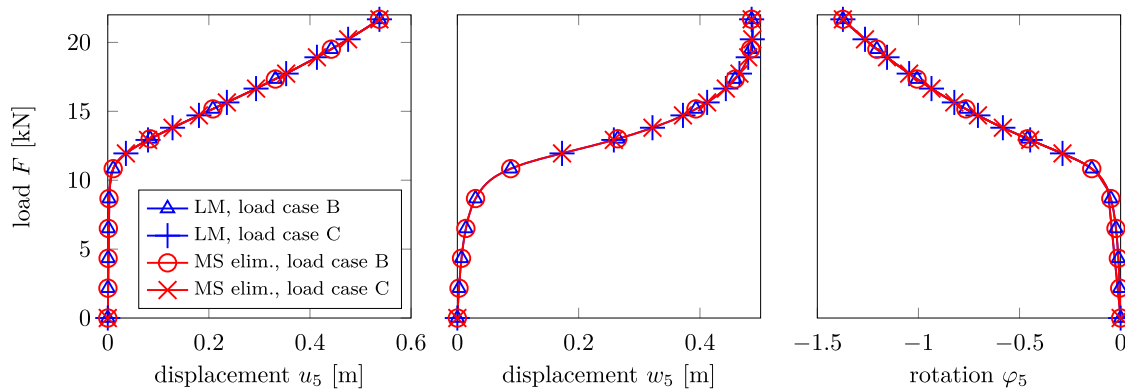


Fig. 16 Coupled rigid and elastic beam: load–displacement diagrams of load cases B (Neumann BC) and C (Dirichlet BC)

A were computed with both the new method as well as Lagrange multipliers for reference. For the computation, 80 load steps are used. The resulting load–displacement diagrams are depicted in Fig. 17. They illustrate the strong nonlinearity of the system behavior. The master–slave elimination leads to the same results as the Lagrange multipliers.

To illustrate the strong nonlinearity more clearly and to demonstrate the suitability of the proposed master–slave elimination in the nonlinear regime, specific deformations are looked more closely at. The true to scale deformation computed with the master–slave elimination for the maximal load $F_{\max} = 100$ kN is depicted at the left side of Fig. 18 and for the minimal load $F_{\min} = -100$ kN at the right side of Fig. 18. Here, the large displacements and rotations and, more spe-

cific, the large displacement discontinuity at the shear-force release can be seen.

In the context of such strong nonlinearity, it is important to assess both the robustness as well as the efficiency. Therefore, for load case A the maximal load step size $\Delta \tau_{\max}$ that does not lead to divergence is analyzed for the new method as well as the other methods listed in Table 4. This maximal load step size corresponds to a minimal number of load steps $n_{\text{load}, \min}$. The Newton–Raphson scheme is aborted as soon as the absolute criterion $\|\mathbf{R}\|^{(k)} \leq 10^{-9}$ is satisfied. To obtain an indicator for the efficiency, the accumulated number of all iteration steps $\sum n_{\text{ite}}$ is calculated.

The results of this analysis can be found in Table 5. The multiplier methods exhibit the most robust performance. A minimum of $n_{\text{load}, \min} = 14$ load steps with a load step size of

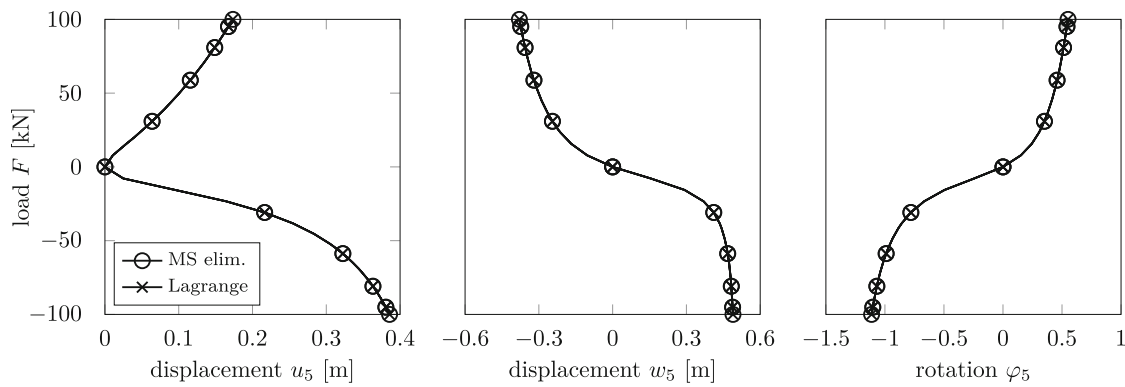


Fig. 17 Coupled rigid and elastic beam: load–displacement diagrams (load case A)

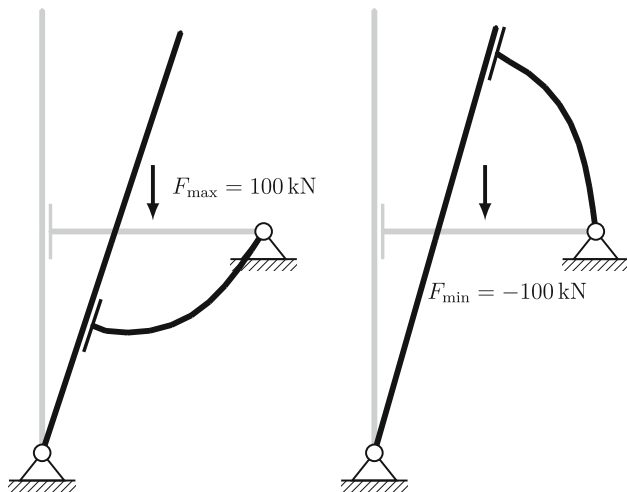


Fig. 18 Coupled rigid and elastic beam: true to scale deformation of system (load case A)

$\Delta\tau_{\max} = \frac{1}{14}$ is needed. For a larger load step size, the solution already diverges in the first load step. In terms of iteration steps, the results indicate that the multiplier methods exhibit also the most efficient performance. In total, $\sum n_{\text{ite}} = 94$ respectively 95 for perturbed Lagrange multipliers iteration steps are needed. The master–slave elimination exhibits a less robust performance. It needs at least $n_{\text{load, min}} = 22$ load steps with a load step size of $\Delta\tau_{\max} = \frac{1}{22}$. In terms of iteration steps, the results indicate that the master–slave elimination exhibits also a less efficient performance. In total, $\sum n_{\text{ite}} = 119$ iteration steps are needed. On the other hand, the dimension of the resulting system is smaller, see Table 1 in Sect. 4.2, leading to reduced time to solve the linearized system, see Sect. 5.4. Additionally, the number of iteration steps can only serve as an indicator since the overall efficiency depends on the computational time of each iteration step too. Due to its heavy dependence on the specific implementation of the constraint methods, the computational time is not further investigated here.

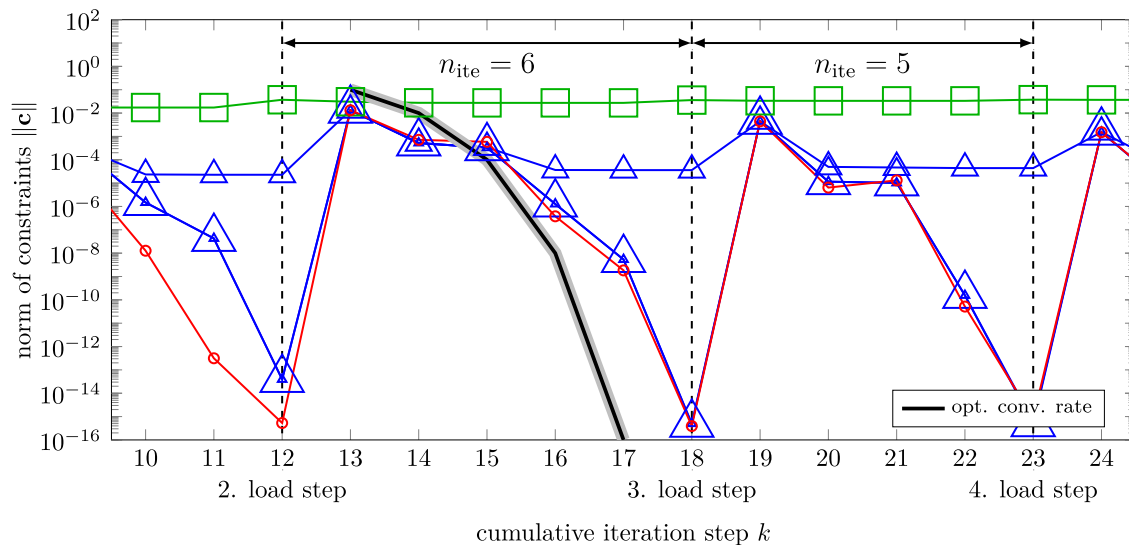
Finally, to analyze the accuracy as well as the convergence behavior specifically for the satisfaction of the constraints, the Euclidean norm of the constraint vector $\|\mathbf{c}\|$ (and not the residual vector $\|\mathbf{R}\|$) within the iterative process of the Newton–Raphson scheme is computed with both the new method as well as with the other methods listed in Table 4 (with the exception of the penalty method with $\beta = 10$ and $\beta = 10^6$). It is depicted in Fig. 19 (representative for 2 of 22 load steps). The constraint norm hits machine precision for master–slave elimination as well as for the Lagrange multiplier and the augmented Lagrange multiplier method. For the penalty method with $\beta = 10^3$, the constraint norm reaches a threshold of a magnitude of 10^{-2} . Similarly, the perturbed Lagrange multiplier method leads to minimal constraint norm of 10^{-5} .

We have also examined and compared the convergence behavior based on the residual vector $\mathbf{R}$ and the unmodified residual vector $\mathbf{R}^*$ (Eq. (2)) exemplary for the 3. of the 22 load steps for all methods listed in Table 4 (with the exception of the penalty method with $\beta = 10$ and $\beta = 10^6$). The constraint norm $\|\mathbf{c}\|$ is depicted over the residual norm $\|\mathbf{R}\|$ on the left side of Fig. 20 and over the norm of the unmodified residual vector $\|\mathbf{R}^*\|$ on the right side of Fig. 20. In this load step, the constraint norm $\|\mathbf{c}\|$ changes very little for the penalty and the perturbed Lagrange multiplier method. For the augmented Lagrange multiplier, the Lagrange multiplier and the master–slave elimination method, the constraint norm $\|\mathbf{c}\|$ and residual norm $\|\mathbf{R}\|$ decreases with the same speed. This shows, that the underlying mechanical problem is strongly influenced by the constraints. The analysis of the unmodified residual norm $\|\mathbf{R}^*\|$ also stresses this conclusion because the order of magnitude of the unmodified residual norm $\|\mathbf{R}^*\|$ changes very little for all methods. Therefore, it is of no importance and should not be considered in the analysis of constrained problems. This also means that it is computationally unreasonable to first solve for the mechanical equations and then modify the solution to incorporate the constraints. The same holds for the opposite strategy. Instead,

Table 5 Robustness and efficiency for rigid beam with nonlinear shear-force release (load case A)

Method	$n_{\text{load, min}}$	$\Delta\tau_{\text{max}}$	$\sum n_{\text{ite}}$
Lagrange multiplier	14	$\frac{1}{14}$	94
Perturbed Lagrange multiplier, $\epsilon = 10^{-6}$	14	$\frac{1}{14}$	95
Augmented Lagrange multiplier, $\beta = 10$	14	$\frac{1}{14}$	94
Penalty method, $\beta = 10$	—*	—*	—*
Penalty method, $\beta = 10^3$	22	$\frac{1}{22}$	122
Penalty method, $\beta = 10^6$	124	$\frac{1}{124}$	625
Master–slave elimination	22	$\frac{1}{22}$	119

*No load size with convergence could be obtained

**Fig. 19** Coupled rigid and elastic beam: convergence behavior of the constraints for 2 of 10 load steps (load case A); for legend, see Table 4

the mechanical equations and the constraints equations have to be solved simultaneously.

The major results of this numerical example can be summarized and interpreted as follows: In the master–slave elimination, it is possible to apply Neumann boundary conditions as well as homogenous and inhomogeneous Dirichlet boundary conditions for dofs involved in constraints. Additionally, the master–slave elimination is as accurate as the Lagrange multiplier method and satisfies the constraints exactly, as analytically expected. In contrast to the other numerical examples presented in this paper, in this example, the master–slave elimination is slightly less robust than the multiplier methods.

6.4 Shearing of slender plane structure with rigid parts

The aim of the fourth example is to evaluate the proposed master–slave elimination scheme with respect to its impact on the condition number and the robustness in the regime of a large number of constraints. Additionally, the dimension of

the resulting linear systems of all methods listed in Table 4 are compared. For this purpose, we analyze a slender plane elastic structure under shear condition (Young’s modulus E , Poisson’s ratio ν and thickness t), see Fig. 21. It is clamped at the bottom and supported by a sliding clamping at the top. The system is loaded at the top with load F . All geometry and stiffness data are stated in Fig. 21.

To analyze the impact of the number of constraints, the middle part of the structure, marked with dark gray in Fig. 21, is subsequently modified to be rigid by introducing constraints. Five different ratios of the rigid part of the domain, indicated by parameter α , are analyzed: $\alpha = \{0\%, 10\%, 20\%, 40\%, 80\%\}$, see Fig. 21.

For the numerical simulation, the system is discretized with 4×40 geometrically nonlinear quadrilateral EAS elements with 4 element nodes and 7 internal variables, see left side of Fig. 21, resulting in a total of $n_{\text{dof}} = 410$ dofs. The load F is applied as a uniform distributed load at the top edge. The load is increased incrementally in 10 steps.

Two variants to model the rigid part of the structured mesh are analyzed: First, in model A, the rigidity is modeled purely

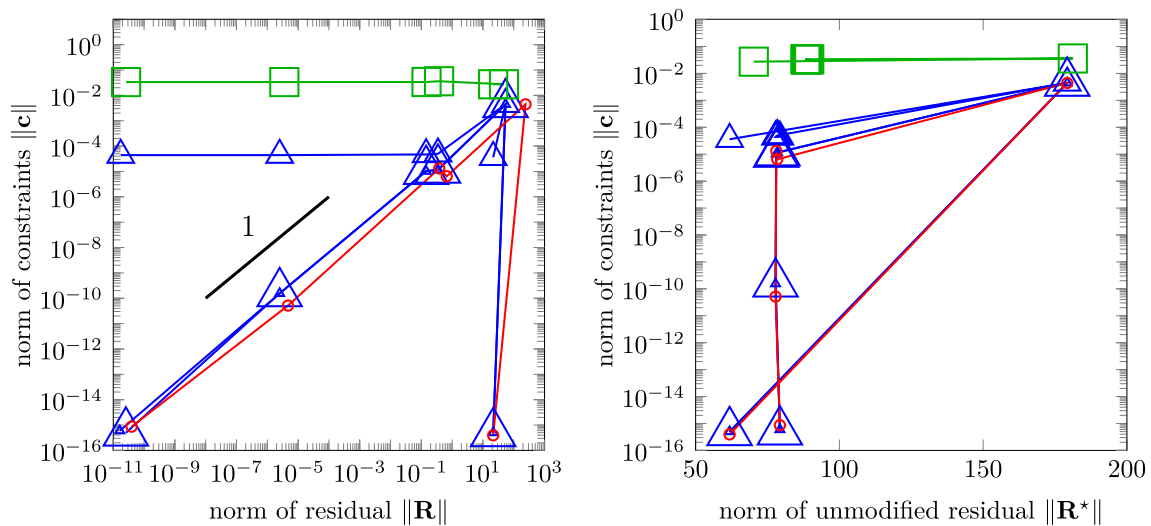


Fig. 20 Coupled rigid and elastic beam: constraint norm over modified and unmodified residual norm (load case A); for legend, see Table 4

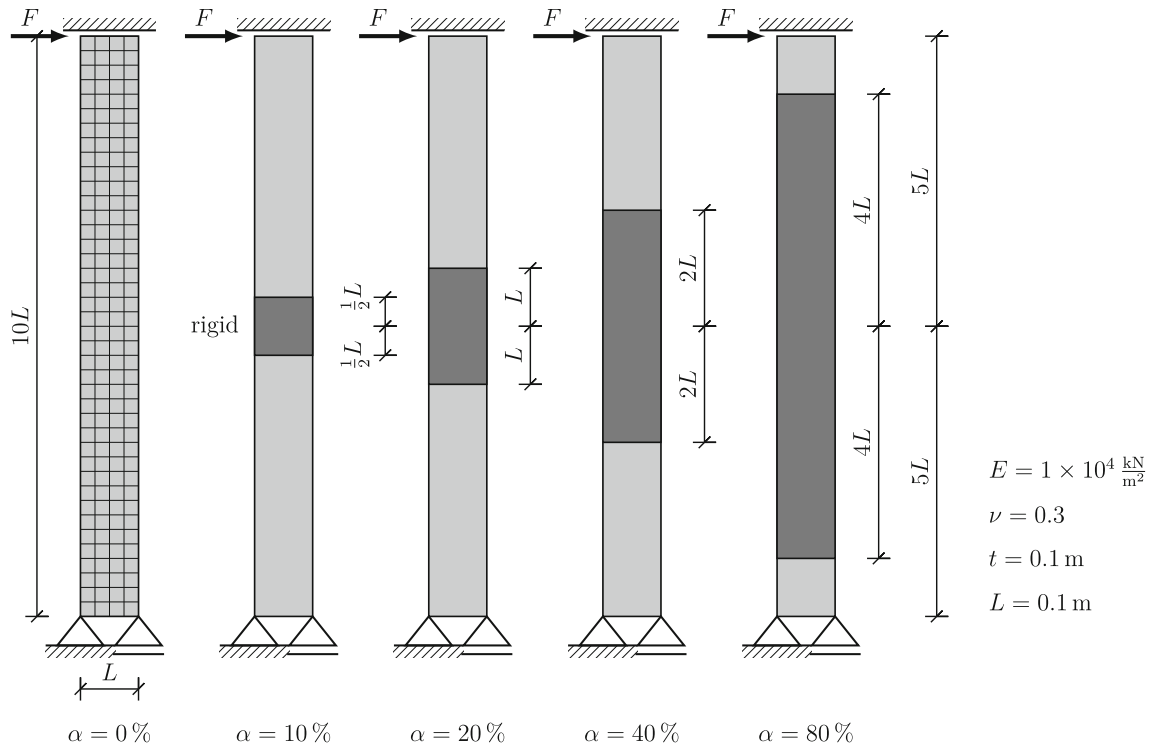


Fig. 21 Shear test: geometry, parameters and mesh

by nonlinear distance constraints $\Delta \ell_{ik} = 0$, see remarks in Sect. 6.1. Here, the nodes of each edge in the finite element mesh is coupled with distance constraints and, additionally, diagonal distance constraints are considered for the bottom and left element rows of the FE-mesh, which act analogously to a bracing (visualized by red lines at the left side of Fig. 22). Second, in model B, the rigidity is modeled by a combination of nonlinear distance constraints $\Delta \ell_{ik} = 0$ and the following

nonlinear angular constraint

$$c = \Delta \theta_{ijk} = \theta_{ijk}(u_i, w_i, u_j, w_j, u_k, w_k) - \Theta_{ijk} = 0 \quad (47)$$

with inclined angle Θ_{ijk} in reference configuration and inclined angle θ_{ijk} in current configuration, depending nonlinearly on the horizontal and vertical displacements u and w of the node triple i, j and k . The associated Jacobian and Hessian of the constraint Eq. (47) were derived and the results are

Fig. 22 Shear test: rigid domain modeled solely by nonlinear distance constraints (model A) and by a combination of nonlinear distance and angular constraints (model B)

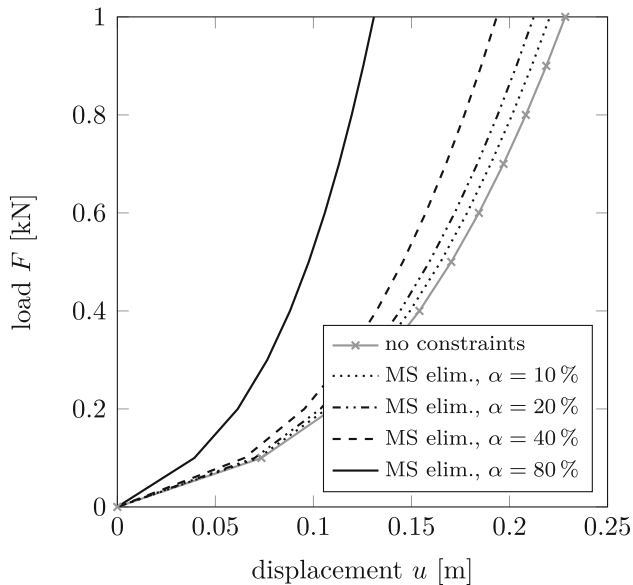
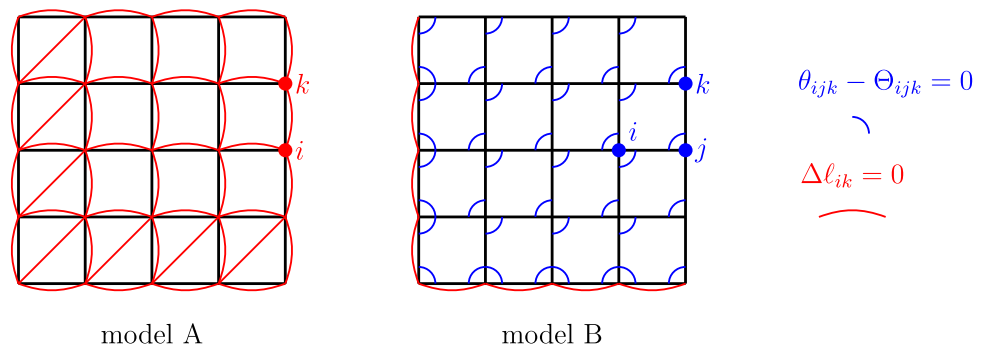


Fig. 23 Shear test: load displacement-diagram for different parameters α (model A)

stated in “Appendix D.4”. In model B, first, the change of the angle of two opposing corners of each finite element is prevented by angular constraints and, second, additional angular constraints are inserted for the bottom and left element rows, which prevent shearing (visualized by blue lines at the right side of Fig. 22). Third, distance constraints are inserted for the bottom and left edges to prevent a change in width and height of the domain (visualized by red lines at the right side of Fig. 22). The connectivity of the constraints is modeled deliberately in such a way that the constraints are not redundant. Both models exhibit the same number of constraints n_c that depends on the parameter α : $n_c|_{\alpha=0\%} = 0$; $n_c|_{\alpha=10\%} = 46$; $n_c|_{\alpha=20\%} = 86$; $n_c|_{\alpha=40\%} = 166$; $n_c|_{\alpha=80\%} = 326$. The constraints are fully coupled (type 3), see Sect. 5.3.

It has to be mentioned that the rigid part can be modeled simply by constraints that directly couple the lower edge of the upper elastic part with the upper edge of the lower elastic part. Such a model exhibits fewer number of finite elements and constraints. However, this model is not used here for

two reasons: First, even in practice they are reasons to not remove the rigid part, e.g. to enable certain areas of the FE-mesh to be modeled once as rigid and once as elastic during the simulation without the need of re-meshing. Secondly, in this example, the worst case for the master–slave elimination of many fully coupled constraints (type 3), see Sect. 5.3, is examined deliberately.

During the complete FE-simulation for all methods, we observe a quadratic rate of convergence in the Newton–Raphson scheme if not stated otherwise.

The numerical results are depicted in Figs. 23, 24, 25, 26 and 27. To study the influence of the parameter α on the deformation of the plane structure, the load–displacement curves for model A were computed with the proposed master–slave elimination scheme and depicted in Fig. 23. The curves show that with an increasing parameter α , the system becomes stiffer, as expected.

In Fig. 24, the deformed meshes for a load $F = 1$ kN and additionally the vertical constraint force C_y computed with the proposed master–slave elimination scheme are depicted as well. Here, the increase in stiffness due to the increasing parameter α can be observed too. The constraint forces are concentrated at the interface between the rigid and the elastic domain. Inside the rigid domain, the constraint forces are 0. A larger parameter α leads to larger constraint forces.

Since it is well known that certain constraint methods like the penalty method have a large influence on the condition number κ of the resulting matrix, the condition number is analyzed more in detail. It is an indicator for the accuracy of the solution of the linear system $\mathbf{A} \mathbf{x} = \mathbf{b}$. The order of magnitude k of $\kappa \approx 10^k$ is an approximation for the loss of accuracy of the solution $\mathbf{x}$ compared to the right hand side $\mathbf{b}$ [14]. Therefore, we computed the condition number κ ¹² of the resulting matrix $\mathbf{K}_{\text{red}}$ of the proposed master–slave elimination scheme, see Eq. (37), of a different number of constraints n_c . The results are depicted for different parameters α using constraint modeling A in Fig. 25. Here, the strong

¹² An approximation of κ was computed using the mathematical function LAPACK_E_dgecon provided by the Math Kernel Library (MKL) by Intel [25].

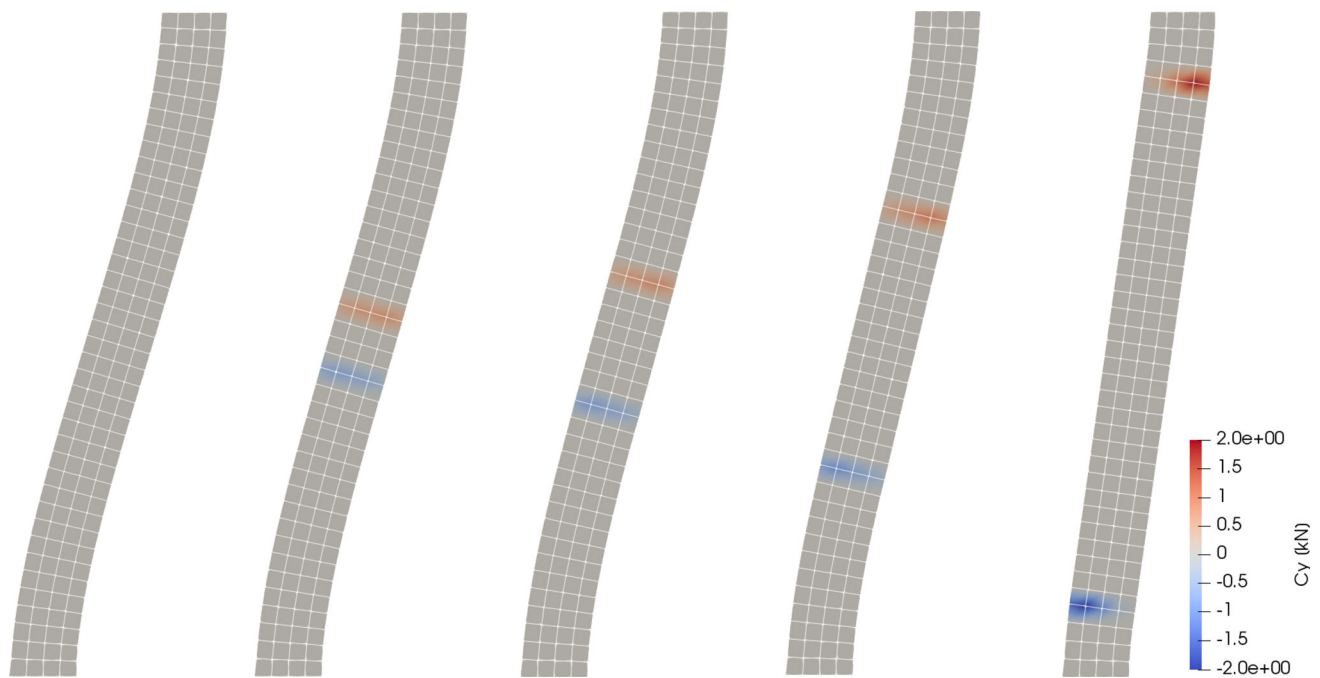


Fig. 24 Shear test: true-to-scale deformed configurations for different parameters α with constraint force C_y (model A)

impact on the condition number κ can be seen. It is increased by two orders of magnitude due to the master–slave elimination. The parameter α and, thus, the number of constraints has merely a minor influence on κ .

To study the influence of the constraint method on the condition number κ of the resulting matrix, the condition number was computed for a parameter $\alpha = 80\%$ with model A using all methods listed in Table 4. The results are depicted in Fig. 26. Here, the strong influence of the method on the condition number κ can be obtained. For the penalty method, the well known increase of the condition number due to an increasing penalty factor can be observed. A larger penalty factor β leads to a larger condition number κ . The augmented Lagrange multipliers lead to a similar result as the penalty method with $\beta = 10$. The Lagrange multipliers and the perturbed Lagrange multipliers have approximately the same condition number as the unconstrained system. In contrast, the master–slave elimination exhibits a condition number two orders of magnitude larger.

Additionally, all these results are compared with those obtained with model B. The load–displacement curves are depicted on the left side of Fig. 27 for a parameter $\alpha = 80\%$. The Lagrange multiplier, the augmented Lagrange multiplier and the penalty methods diverge in the first load step (therefore, they are not depicted). This illustrates the sensitivity of these methods for this example. For model B, the master–slave elimination scheme and the perturbed Lagrange multiplier converge quadratically and produce the

same results as for model A. The associated condition number κ is depicted on the right side of Fig. 27, the master–slave elimination increases the condition number by approximately 2 orders of magnitude in comparison to the unconstrained system. Here, the same behavior as for model A can be observed. For model B, the perturbed Lagrange multipliers increases the condition number by approximately 1.5 orders of magnitude in comparison to the unconstrained system. In contrast to the master–slave elimination, the constraint model has an impact on the condition number for the perturbed Lagrange multipliers. We suppose this is due to the use of the same perturbation factor for the two different constraint types $\Delta\ell_{ik} = 0$ and $\Delta\theta_{ijk} = 0$ although their physical dimension differs. A different weighting of the constraints could lead to the same condition number for model A and B.

To examine the influence of the parameter α and, thus, the number of constraints on the dimension of the problem, the dimension of the resulting matrix n_{dim} for the both the new method as well as with the other methods listed in Table 4 is computed. The number of degrees of freedom is $n_{\text{dof}} = 410$ identical for all cases. The results are summarized in Table 6. Here, the clear advantage of the master–slave elimination compared to penalty method but especially to multiplier methods in form of the reduction of the dimension can be seen. The dimension reduction is significant for the master–slave elimination and increases with increasing parameter α .

Fig. 25 Shear test: condition number of resulting matrix for different parameters α (model A)

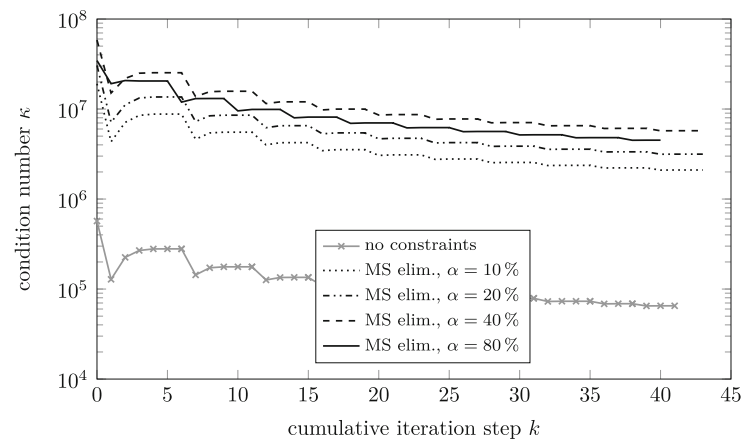


Fig. 26 Shear test: condition number of resulting matrix for different methods (model A); for legend, see Table 4

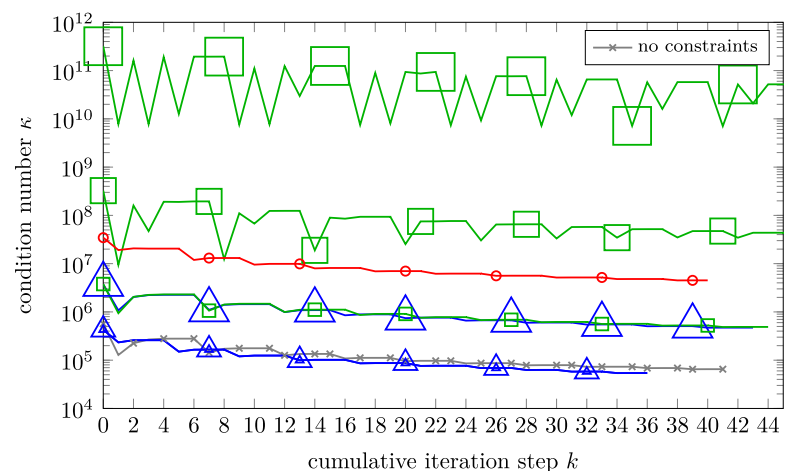


Table 6 Shear test: dimension of resulting system n_{dim} depending on the method and parameter α

Method	$\alpha = 10(\%)$		$\alpha = 20(\%)$		$\alpha = 40(\%)$		$\alpha = 80(\%)$	
	n_{dim}	$\frac{n_{\text{dim}}}{n_{\text{dof}}} (\%)$	n_{dim}	$\frac{n_{\text{dim}}}{n_{\text{dof}}} (\%)$	n_{dim}	$\frac{n_{\text{dim}}}{n_{\text{dof}}} (\%)$	n_{dim}	$\frac{n_{\text{dim}}}{n_{\text{dof}}} (\%)$
PM	410	100	410	100	410	100	410	100
LM, ALM, PLM	456	111	496	121	576	140	736	180
MS. elim	364	89	324	79	244	60	84	20

The major results of this numerical example can be summarized and interpreted as follows: The master–slave elimination leads to a moderate increase of the condition number compared to a system without constraints. Additionally, the master–slave elimination and the perturbed Lagrange multipliers are robust with the respect of the modeling of the rigidity but the condition number is only independent of the model for the master–slave elimination. Finally, the example also illustrates the beneficial dimension reduction of the master–slave method due to the elimination.

6.5 Stability analysis of thin-walled beam

The aim of the fifth and last example is to evaluate the proposed master–slave elimination scheme in the context of post-critical behavior and stability problems. For this purpose, we analyze a thin-walled elastic cantilever beam with a U-shaped cross-section (Young’s modulus E , Poisson’s ratio ν , height H , width B , length L and thickness t). Two different variants of the beam are investigated: Model A, which is the baseline model without additional plates, see left side of Fig. 28, and model B, in which the beam is stiffened with four additional plates, see middle and right side of Fig. 28.

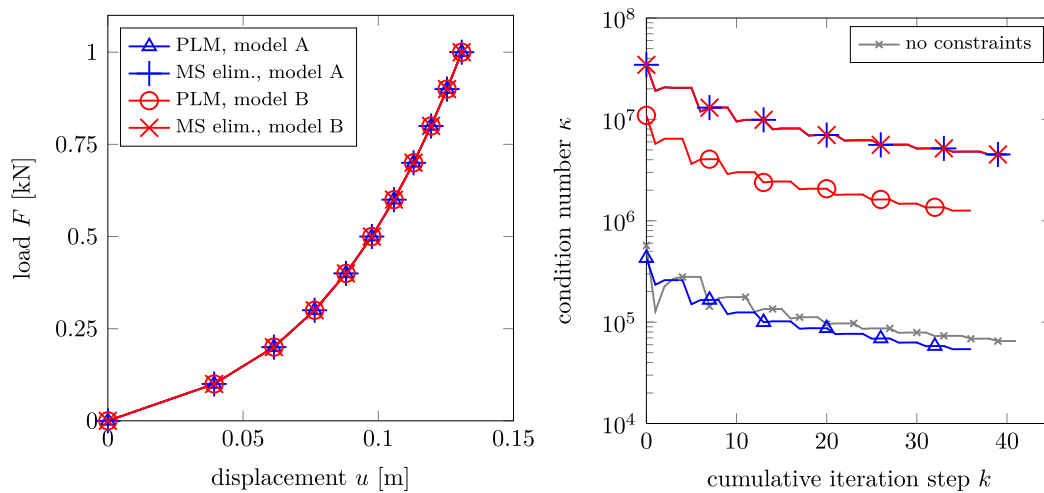


Fig. 27 Shear test: load–displacement diagrams (left) and condition number of resulting matrix (right) (model A and B)

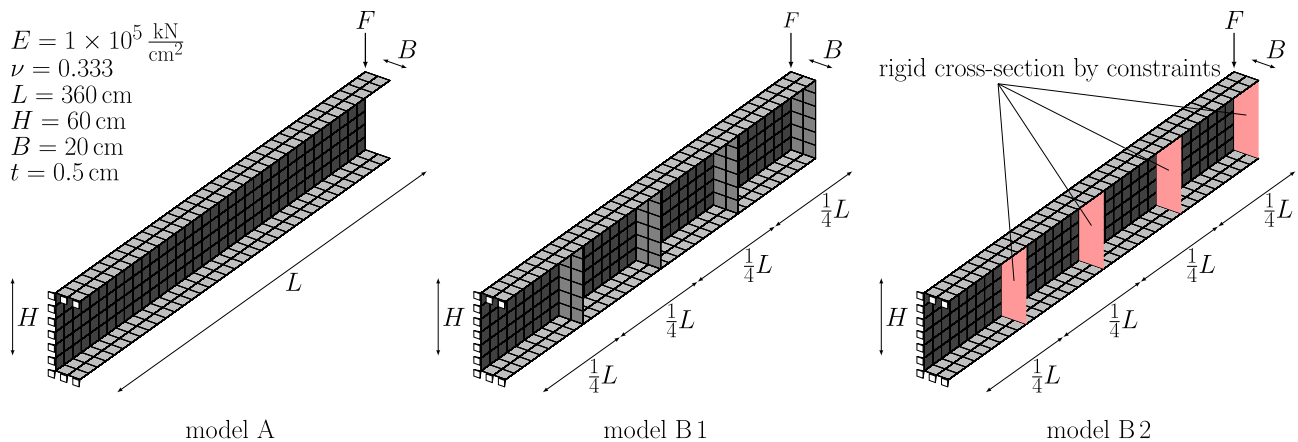


Fig. 28 Shell stability analysis: geometry, parameters and coarsest mesh with 10×36 elements. Three different models are investigated

The beam is clamped at the left side. The structure is loaded at the end of the cantilever at the top of the cross-section with load F . All geometry and stiffness data are stated in Fig. 28.

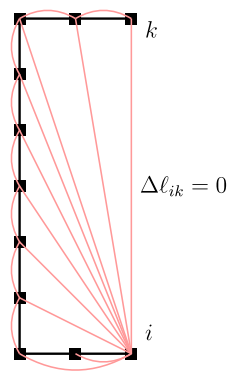
For the numerical simulation, the system is discretized with 10×36 , 15×54 and 20×72 geometrically nonlinear Mindlin–Reissner finite shell elements with finite rotations (based on [41]) using the MITC approach by Bathe and Dvorkin [7] to prevent shear locking, see left side of Fig. 28 for coarsest mesh. Based on a convergence study, the finest mesh with 20×72 elements was chosen for the subsequent calculations. For the analysis of the post-critical behavior, the load is applied by means of a displacement control. The vertical displacement w is increased incrementally in 100 steps to a maximum of $w = 20 \text{ cm}$. For the nonlinear stability analysis, the load is applied by means of a load control. A base load of $F = 100 \text{ kN}$ is used. For the stability criterion, the number of non-positive eigenvalues of the effective matrices, see corresponding equations in Table 4, at the end

of each equilibrium iteration was observed. The eigenproblem was solved using the FEAST eigensolver [32] available in the MKL [25]. For the challenges arising in using this type of solver in the context of multiplier methods, see Sect. 4.2. For identification of the instability point and the computation of the critical load F_{crit} , the one-sided bisection method was used. It is aborted as soon as the criterion $\|\Delta F\|/F < 10^{-5}$ is satisfied.

Two variants to model the additional plates are analyzed: First, in model B1, additional shell elements of the same material and thickness are introduced, see middle of Fig. 28. Second, in model B2, the plates are modeled as rigid by introducing distance constraints in the corresponding cross-section, see right side of Figs. 28 and 29, without introducing additional finite elements.

The numerical results are depicted in Fig. 30. To verify the baseline model A and to analyze the impact of the additional plates, the load–displacement curves were com-

Fig. 29 Shell stability analysis: pattern of distance constraints for rigid cross-section



puted for all three models, see left side of Fig. 30. All three models result in different load–displacement curves. The solution for the baseline model A lies in good agreement to the reference solution from [10] which verifies the baseline model. The load–displacement curves of model B 1 and model B 2 show a stiffer behavior than model A due to the additional plates. In turn, model B 1 and model B 2 result in different load–displacement curves. This is due to the different stiffening behavior of the additional finite elements and the constraints. While the additional finite elements are rigidly coupled to the beam with respect to displacements and rotations, the constraints only affect the displacements of the beam. Modeling the plates with additional finite elements allows for deformation of the plates, for example buckling of the plates, whereas modeling the plates with constraints allows for neither in-plane nor out-of-plane deformation. For model B 2, the load–displacement curve was computed with both the new method as well as with the other methods listed in Table 4, see right side of Fig. 30. The results for the master–slave elimination, the Lagrange multipliers and the augmented Lagrange multipliers lie in good agreement. The solution for the penalty method and the perturbed Lagrange multipliers exhibit visible differences. For the penalty method with a penalty factor of $\beta = 10$, the solution diverges for a displacement of $w = 14.6$ cm, and for $\beta = 10^3$ the solution diverges for a displacement of $w = 13.2$ cm.

To analyze the impact and to demonstrate the ease of use of the master–slave elimination in stability analysis, a nonlinear stability analysis and search for the point of instability with the bisection method was performed for model B 2 with both the new method as well as with the other methods listed in Table 4, see right side of Fig. 30. In addition, this analysis was performed for model B 1. The instability point is a limit point. The resulting critical loads are summarized in Table 7. The eigenvectors related to F_{crit} for model B 1 and model B 2 computed with the master–slave elimination are depicted in Fig. 31.

The results for the critical loads for model B 1 and B 2 differ due to the same reasons mentioned for the load displacement curve. For model B 2, the results for the

master–slave elimination lies in good agreement with the augmented Lagrange multipliers and the penalty method for large penalty factors. For the Lagrange multipliers, the linear equation solver leads to a termination of the calculation for a load of $F = 116$ kN near the actual instability load. This is due to an error in the solution procedure for the linear system caused by finding a pivot element equal to zero.

The major results of this numerical example can be summarized and interpreted as follows: The master–slave elimination can be used for complex nonlinear structures. Additionally, the method can be embedded in nonlinear stability analysis, the bisection method for finding the point of instability and the computation of the corresponding eigenproblem without any modifications, in contrast to the multiplier methods in which the indefinite matrix has to be considered. For the nonlinear stability analysis, the results lie in agreement with existing constraint methods. Besides, the master–slave elimination is capable of capturing the post-critical behavior correctly.

7 Conclusions

Compared to other constraint methods, the master–slave elimination method leads to a significant reduction of the dimension of the resulting linear system which is of great advantage, especially when a large number of constraints have to be considered. However, the significant disadvantage of the existing master–slave elimination schemes in the literature is their limitation to linear constraints.

In this paper, we presented an extension of the master–slave elimination to consider arbitrary smooth nonlinear constraints. The derivation of the method was deliberately performed in such a manner that allows for a profound comparison with other constraint methods. This comparison can take place on the basis of the linear system to be solved in the individual constraint methods (resulting from the consistent linearization). On the basis of the theoretical analysis, it could be shown that the proposed master–slave elimination scheme exhibits a reduced computational complexity compared to other constraint methods.

It could also be shown that the new method can be implemented in a very general form so that a new constraint can be added very easily as a new item in a “constraint library”, without the need for a case-by-case programming.

Novel benchmark tests with analytical and reference solutions were introduced to assess the performance of the constraint methods in general. These tests were deliberately designed to analyze a broad spectrum of properties. Different classes of constraints were used both small and large academic examples. This enables us to make generalized statements. With these tests, the performance of the new method was compared with the penalty and multiplier meth-

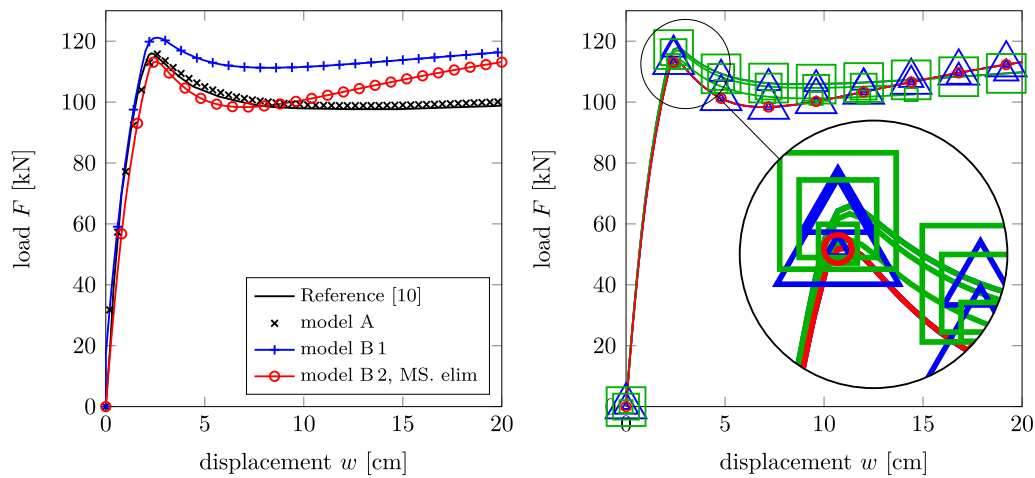


Fig. 30 Shell stability analysis: load–displacement diagram; for legend, see Table 4

Table 7 Critical load for model B 1 and model B 2

Model and method	$F_{\text{crit.}}$ [kN]
Model B 1	121
Model B 2, Lagrange multiplier	–
Model B 2, perturbed Lagrange multiplier, $\epsilon = 10^{-6}$	112
Model B 2, augmented Lagrange multiplier, $\beta = 10$	117
Model B 2, penalty method, $\beta = 10$	114
Model B 2, penalty method, $\beta = 10^3$	117
Model B 2, penalty method, $\beta = 10^6$	118
Model B 2, master–slave elimination	118

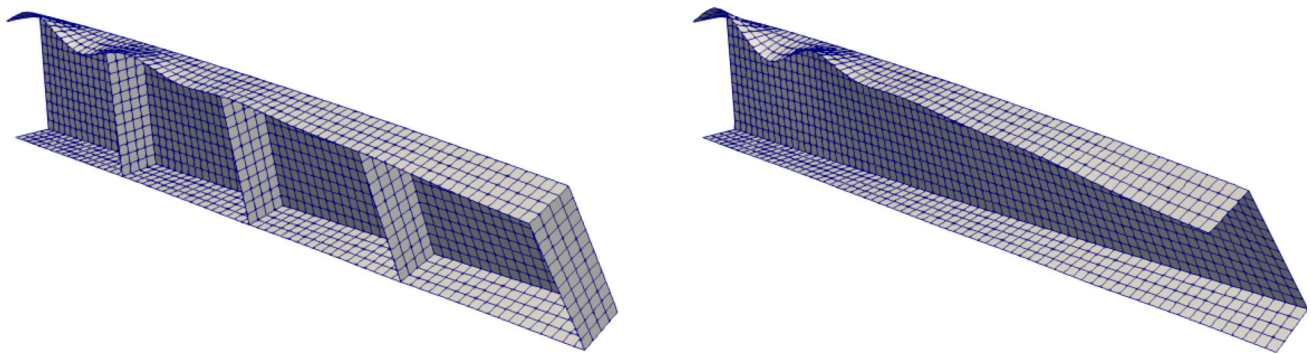


Fig. 31 Shell stability analysis: eigenvector related to $F_{\text{crit.}}$ for model B 1 (left) and model B 2 (right) computed with the master–slave elimination

ods. This comparison shows the following properties of the new method:

- It can deal with arbitrary smooth decoupled and coupled nonlinear multi-point constraints.
- It exhibits quadratic convergence which verifies the derived linearization in Eq. (36), see Sect. 6.1 for a corresponding graphical representation.
- It satisfies the constraints exactly within the limits of the computational accuracy, see Sects. 6.1, 6.2 and 6.3.
- It is accurate. It produces the same results as the analytical solutions (if available), respectively the Lagrange multipliers, see Sect. 6.1, 6.2 and 6.3.
- It reduces the dimension of the problem proportionally to the number of constraints, see Sect. 6.4. This is a great advantage, especially compared to Lagrange multipliers

that increase the dimension proportionally to the number of constraints.

- It is robust. For certain examples, the multiplier methods performed better, see Sect. 6.3. However, the master–slave elimination was able to handle examples that lead to divergence for the Lagrange multiplier, augmented Lagrange multiplier and the penalty method, see Sect. 6.4.
- It can handle both Neumann and Dirichlet boundary conditions applied to dofs that are involved in the constraints, see Sect. 6.3.
- It leads to a moderate increase of the condition number κ compared to a system without constraints, whereas the influence of the number of constraints is neglectable. In contrast to other constraint methods, its condition number is also independent of the choice of specific constraints, see Sect. 6.4.

All presented results are derived in the context of the FEM. However, the proposed method can also be applied in other discretization methods.

The choice of an appropriate set of slave dofs, required for the master–slave elimination scheme, was not discussed in this paper. The same holds for redundant or contradictory constraints. Both aspects will be discussed in a follow-up paper [11]. In the current paper, we were focusing on scleronomic constraints. It is of great interest to extend the proposed new master–slave elimination for rheonomic and non-holonomic constraints and thus dynamic problems. Additionally, the method in the presented version is restricted to smooth constraints. Finally, nonlinear multi-point constraints have not yet been discussed in the literature in the context of path following algorithms and linear stability analysis. Specific adjustments of the proposed new master–slave elimination scheme are needed for these methods. These extensions are a part of our ongoing research.

A Identity of perturbed Lagrange multipliers and penalty method

As indicated in Sect. 3.5, the identity of perturbed Lagrange multipliers with elimination of the multipliers with the penalty method is shown hereafter. For this, both the modified function for perturbed Lagrange multipliers and for the penalty method are recalled:

$$W_{\text{PLM}}(\mathbf{V}, \boldsymbol{\lambda}, \epsilon) = W(\mathbf{V}) + \boldsymbol{\lambda}^T \mathbf{c}(\mathbf{V}) - \frac{1}{2} \epsilon \boldsymbol{\lambda}^T \boldsymbol{\lambda};$$

$$W_{\text{PM}}(\mathbf{V}, \beta) = W(\mathbf{V}) + \frac{1}{2} \beta \mathbf{c}(\mathbf{V})^T \mathbf{c}(\mathbf{V}) \quad (48)$$

If the Lagrange multipliers $\boldsymbol{\lambda}$ are approximated by the following equation

$$\boldsymbol{\lambda} = \frac{1}{\epsilon} \mathbf{c} = \beta \mathbf{c} \quad \text{with} \quad \frac{1}{\epsilon} \equiv \beta \quad (49)$$

the second term of the modified functions Eqs. (48)₁ and (48)₂ are equal:

$$\begin{aligned} \boldsymbol{\lambda}^T \mathbf{c} - \frac{1}{2} \epsilon \boldsymbol{\lambda}^T \boldsymbol{\lambda} &= \boldsymbol{\lambda}^T \left(\mathbf{c} - \frac{1}{2} \epsilon \boldsymbol{\lambda} \right) \\ &= \beta \mathbf{c}^T \mathbf{c} - \frac{1}{2} \frac{1}{\beta} \beta \mathbf{c}^T \beta \mathbf{c} \equiv \frac{1}{2} \beta \mathbf{c}^T \mathbf{c} \end{aligned} \quad (50)$$

With the equivalence of the second terms of the function shown in Eq. (50), the underlying functions for the minimization Eqs. (48)₁ and (48)₂ themselves are equal.

B Symmetry of the reduced stiffness matrix

As indicated at the end of Sect. 4.1, the symmetry of the reduced stiffness matrix $\mathbf{K}_{\text{red}}$ defined in Eq. (36) under the assumption of the symmetry of the tangential stiffness matrix of the unconstrained problem $\mathbf{K}_T$ is proven hereafter. Due to Schwarz's theorem the Hessian of the constraints $\mathbf{H}_i$ is symmetric too. To verify the symmetry of $\mathbf{K}_{\text{red}}$, each term of the matrix is considered separately in the following:

$$\begin{aligned} \mathbf{K}_{\text{red}} &= \mathbf{K}_{mm} - \sum_{i=1}^{n_c} (\mathbf{H}_{mm,i} Y_i) + \mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{sm,i} Y_i) \\ &\quad - \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{sm} - \\ &\quad \left(\mathbf{K}_{ms} - \sum_{i=1}^{n_c} (\mathbf{H}_{ms,i} Y_i) + \mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{ss,i} Y_i) \right. \\ &\quad \left. - \mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{ss} \right) \mathbf{G}_s^{-1} \mathbf{G}_m \\ &= \underbrace{\mathbf{K}_{mm}}_{\text{term 1}} + \underbrace{\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{ss} \mathbf{G}_s^{-1} \mathbf{G}_m}_{\text{term 2}} - \underbrace{\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{sm} - \mathbf{K}_{ms} \mathbf{G}_s^{-1} \mathbf{G}_m}_{\text{term 3}} \\ &\quad + \underbrace{\sum_{i=1}^{n_c} (\mathbf{H}_{mm,i} Y_i)}_{\text{term 4}} - \underbrace{\mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{ss,i} Y_i) \mathbf{G}_s^{-1} \mathbf{G}_m}_{\text{term 5}} \end{aligned} \quad (51)$$

$$+ \underbrace{\mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{sm,i} Y_i) + \sum_{i=1}^{n_c} (\mathbf{H}_{ms,i} Y_i) \mathbf{G}_s^{-1} \mathbf{G}_m}_{\text{term 6}}$$

For the matrix $\mathbf{K}_{\text{red}}$ to be symmetric, the individual terms 1 to 6 must be symmetric in turn.

Term 1 in Eq. (51) $\mathbf{K}_{mm}$ is symmetric because $\mathbf{K}_T$ is symmetric. Term 2 in Eq. (51) $\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{ss} \mathbf{G}_s^{-1} \mathbf{G}_m$ is symmetric because the arbitrary matrix product $\mathbf{A}^T \mathbf{B} \mathbf{A}$ is symmetric if $\mathbf{B} = \mathbf{K}_{ss}$ is symmetric. This is the case because $\mathbf{K}_T$ is symmetric. Term 3 in Eq. (51) $-\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{sm} - \mathbf{K}_{ms} \mathbf{G}_s^{-1} \mathbf{G}_m$ is symmetric. Due to the symmetry of $\mathbf{K}_T$, the first part of the third term can be rewritten to

$$\begin{aligned} \mathbf{A} &:= -\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{sm} = -\mathbf{G}_m^T \mathbf{G}_s^{-T} \mathbf{K}_{ms}^T \\ &= -(\mathbf{K}_{ms} \mathbf{G}_s^{-1} \mathbf{G}_m)^T \end{aligned} \quad (52)$$

so it is the transposed of the second part of the third term, resulting in a sum of the form $\mathbf{A} + \mathbf{A}^T$, which is always symmetric. Term 4 in Eq. (51) $\sum_{i=1}^{n_c} (\mathbf{H}_{mm,i} Y_i)$ is symmetric because of the symmetry of the Hessian $\mathbf{H}_i$. Term 5 in Eq. (51) $-\mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{ss,i} Y_i) \mathbf{G}_s^{-1} \mathbf{G}_m$ is symmetric because the arbitrary matrix product $\mathbf{A}^T \mathbf{B} \mathbf{A}$ is symmetric if $\mathbf{B} = \mathbf{H}_{ss,i}$ is symmetric. This is the case because $\mathbf{H}_i$ is symmetric. Term 6 in Eq. (51) $\mathbf{G}_m^T \mathbf{G}_s^{-T} \sum_{i=1}^{n_c} (\mathbf{H}_{sm,i} Y_i) + \sum_{i=1}^{n_c} (\mathbf{H}_{ms,i} Y_i) \mathbf{G}_s^{-1} \mathbf{G}_m$ is symmetric with the same arguments as for term 3.

Since all terms were shown to be symmetric, $\mathbf{K}_{\text{red}}$ is proven to be symmetric as well.

C Analytical solutions

C.1 Snap-through problem with rigid truss

For the analytical solution of the snap-through problem with rigid truss, the deformed configuration depicted in Fig. 32 is analyzed.

The deformed configuration is uniquely described by the vertical displacement w_2 of node 2 and the horizontal displacement u_3 of node 3. With these quantities, the horizontal displacement u_2 of node 2, the length of the right truss ℓ_{23} , the angles α and β can be computed:

$$\begin{aligned} u_2 &= \sqrt{2L^2 + (L - w_2)^2}; \quad \ell_{23} = \sqrt{(L - u_2 + u_3)^2 + (L - w_2)^2}; \\ \alpha &= \text{asin}\left(\frac{L - w_2}{\sqrt{2}L}\right); \quad \beta = \text{asin}\left(\frac{L - w_2}{\ell_{23}}\right) \end{aligned} \quad (53)$$

The external force F , the constraint force in the left truss C_{12} and the axial force in the right truss N_{23} can be computed

by the equilibrium equations:

$$\begin{aligned} \rightarrow: \quad C_{12} \cos(\alpha) &= N_{23} \cos(\beta) \Rightarrow C_{12} = \frac{\cos(\beta)}{\cos(\alpha)} N_{23} \\ \uparrow: \quad F &= -C_{12} \sin(\alpha) - N_{23} \sin(\beta) \\ \Rightarrow \quad N_{23} &= \left(-\sin(\beta) - \sin(\alpha) \frac{\cos(\beta)}{\cos(\alpha)} \right)^{-1} F \end{aligned} \quad (54)$$

Equations (53) and (54) depend on both the vertical displacement w_2 as well as the horizontal displacement u_3 . However, we aim for a function of the external force solely depending on the vertical displacement $F(w_2)$. To be able to eliminate u_3 , an additional equation has to be introduced in the form of the constitutive law for the axial force in the right truss N_{23} based on Hencky strain measure:

$$N_{23} = EA \ln \left(\frac{\ell_{23}}{\sqrt{2}L} \right) \quad (55)$$

By setting Eqs. (54)₃ and (55) to be equal, the displacement u_3 could be eliminated. No closed solution exists. The resulting equation can be solved numerically for a given vertical displacement w_2 . This results in a clear function for the external load depending on the displacement $F(w_2)$. Here, MATLAB was used for this process.

C.2 Beam with nonlinear shear force release

For the derivation, the solution of the left part is presented at first. There is pure bending with a constant moment of $M/2$ and no shear force,¹³ which allows to derive the following solution by purely geometrical considerations:

$$\begin{aligned} \varphi_2 &= \frac{M}{2EI} L; \\ u_2 &= \frac{2EI \sin(\varphi_2)}{M} - L; \\ w_2 &= (1 - \sin(\varphi_2)) \frac{2EI}{M} \end{aligned} \quad (56)$$

Due to the shear-force release, the rotation at node 2 and 3 is identical ($\varphi_2 = \varphi_3$) and the vertical displacements at node 2 and 3 are antimetric ($w_2 = -w_3$). For the calculation of the horizontal displacement u_3 , the discontinuities in the deformed configuration in the local coordinate system of the force release $\Delta u'$ and $\Delta w'$, see Fig. 12, are needed which are calculated as follows by purely geometrical considerations:

¹³ Here, Timoshenko and Bernoulli beam theory lead to the same results.

Fig. 32 Deformed configuration of the snap-through problem with spring and nonlinear distance constraint

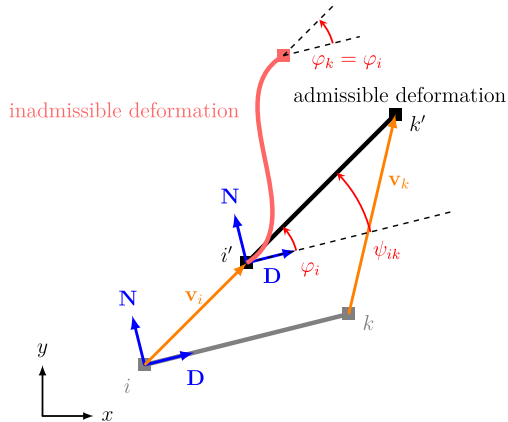
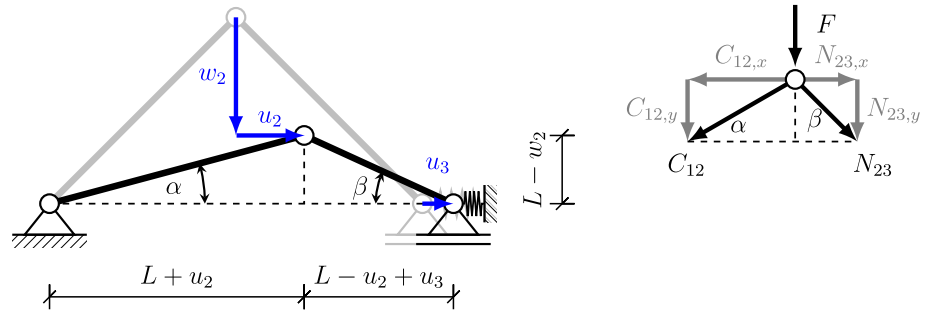


Fig. 33 Third constraint for rigid plane beam

$$\begin{bmatrix} \Delta u' \\ \Delta w' \end{bmatrix} = \begin{bmatrix} 0 \\ 2w_2 \end{bmatrix} = \begin{bmatrix} \cos(\varphi_2) & \sin(\varphi_2) \\ -\sin(\varphi_2) & \cos(\varphi_2) \end{bmatrix} \begin{bmatrix} u_3 - u_2 \\ w_3 - w_2 \end{bmatrix} \quad (57)$$

This leads to the following expression for the horizontal displacement u_3 at the right side of the shear-force release:

$$u_3 = \left(\frac{2EI \sin(\varphi_2)}{M} - L \right) + 4 \tan(\varphi_2) (1 - \sin(\varphi_2)) \frac{EI}{M} \quad (58)$$

For the sake of completeness, the linear equivalent of shear-force release is described by the two linear constraints $\varphi_2 = \varphi_3$ and $u_2 = u_3$.

ties were derived with the help of the symbolic toolbox by MATLAB.

D.1 Inextensibility constraint

The constraint c for the constant distance between two points i and k Eq. (44) is recalled here

$$c(u_i, w_i, u_k, w_k) = \Delta \ell_{ik} = \|\mathbf{X}_i - \mathbf{X}_k\| - \|\mathbf{x}_i - \mathbf{x}_k\| = 0 \quad (59)$$

$$\mathbf{X} \in \mathbb{R}^{n_{\text{dim}}}$$

The associated Jacobian $\mathbf{G}$ is

$$\mathbf{G} = \frac{1}{\Delta \ell_{ik}} \begin{bmatrix} 1 & -1 \end{bmatrix} \Delta \mathbf{x}_{ik}; \quad \mathbf{G} \in \mathbb{R}^{2n_{\text{dim}}} \quad \mathbf{1} \in \mathbb{R}^{n_{\text{dim}} \times n_{\text{dim}}} \quad (60)$$

The associated Hessian $\mathbf{H}$ is

$$\mathbf{H} = \frac{1}{\Delta \ell_{ik}} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} - \frac{1}{\Delta \ell_{ik}^3} \begin{bmatrix} \Delta \mathbf{x}_{ik} \Delta \mathbf{x}_{ik}^T & -\Delta \mathbf{x}_{ik} \Delta \mathbf{x}_{ik}^T \\ -\Delta \mathbf{x}_{ik} \Delta \mathbf{x}_{ik}^T & \Delta \mathbf{x}_{ik} \Delta \mathbf{x}_{ik}^T \end{bmatrix}; \quad \mathbf{H} \in \mathbb{R}^{2n_{\text{dim}} \times 2n_{\text{dim}}} \quad (61)$$

D.2 Nonlinear force release

Here, the second nonlinear constraint c of the force release two points i and k Eq. (45)₂ is recalled:

$$\mathbf{n}_i(\varphi_i)^T \begin{bmatrix} u_i \\ w_i \end{bmatrix} - \mathbf{n}_k(\varphi_k)^T \begin{bmatrix} u_k \\ w_k \end{bmatrix} = 0 \quad (62)$$

The associated Jacobian $\mathbf{G}$ is:

$$\mathbf{G} = \begin{bmatrix} \cos(\varphi_i) & \sin(\varphi_i) & -\sin(\varphi_i)u_i + \cos(\varphi_i)w_i - \cos(\varphi_k) & -\sin(\varphi_k) & \sin(\varphi_k)u_k - \cos(\varphi_k)w_k \end{bmatrix} \quad (63)$$

The associated Hessian $\mathbf{H}$ is:

D Constraint quantities

In this section, the Jacobian and Hessian of the nonlinear constraints handled in this paper are presented. These quan-

$$\mathbf{H} = \begin{bmatrix} 0 & 0 & -\sin(\varphi_i) & 0 & 0 & 0 \\ 0 & 0 & \cos(\varphi_i) & 0 & 0 & 0 \\ -\sin(\varphi_i) & \cos(\varphi_i) & -\cos(\varphi_i)u_i - \sin(\varphi_i)w_i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sin(\varphi_k) \\ 0 & 0 & 0 & 0 & 0 & -\cos(\varphi_k) \\ 0 & 0 & 0 & \sin(\varphi_k) & -\cos(\varphi_k) & \cos(\varphi_k)u_k + \sin(\varphi_k)w_k \end{bmatrix} \quad (64)$$

D.3 Rigid beam

Here, only the third, nonlinear constraint c of the rigid plane beam defined by the points i and k Eq. (46)₃ is recalled

$$c(u_i, w_i, \varphi_i, u_k, w_k) = \varphi_i - \psi_{ik} = \varphi_i - \text{atan2}(d_k^*, d_i^*) = 0 \quad (65)$$

with the two quantities d_k^* and d_i^* that can be defined as follows:

$$\begin{bmatrix} d_i^* \\ d_k^* \end{bmatrix} = \begin{bmatrix} D_x & -D_y \\ D_y & D_x \end{bmatrix} (\mathbf{x}_k - \mathbf{x}_i) \quad (66)$$

with the direction vector $\mathbf{D} = [D_x, D_y]$ illustrated in Fig. 33. Figure 33 also generally illustrates the constraint Eq. (65).

The associated Jacobian $\mathbf{G}$ is

$$\mathbf{G} = \begin{bmatrix} \frac{\Delta_y}{\ell_{ik}^2} & -\frac{\Delta_x}{\ell_{ik}^2} & 1 & \frac{\Delta_y}{\ell_{ik}^2} & -\frac{\Delta_x}{\ell_{ik}^2} & 0 \end{bmatrix} \quad (67)$$

with $\Delta_x := x_i - x_k$; $\Delta_y := y_i - y_k$

The associated Hessian $\mathbf{H}$ is

$$\mathbf{H} = \begin{bmatrix} -\frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & \frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & 0 & \frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & -\frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & 0 \\ \frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & \frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & 0 & -\frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & -\frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & -\frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & 0 & -\frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & \frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & 0 \\ -\frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & -\frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & 0 & \frac{\Delta_x^2 - \Delta_y^2}{\ell_{ik}^4} & \frac{2\Delta_x\Delta_y}{\ell_{ik}^4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (68)$$

D.4 Angular constraint

The constraint c for the constant angle included by three points i , j and k Eq. (47) is recalled here:

$$\begin{aligned} \theta_{ijk} - \Theta_{ijk} &= 0 \\ \theta_{ijk} &= \arccos\left(\frac{(\mathbf{x}_j - \mathbf{x}_i)^T(\mathbf{x}_j - \mathbf{x}_k)}{\|\mathbf{x}_j - \mathbf{x}_i\| \|\mathbf{x}_j - \mathbf{x}_k\|}\right) \\ \Theta_{ijk} &= \arccos\left(\frac{(\mathbf{X}_j - \mathbf{X}_i)^T(\mathbf{X}_j - \mathbf{X}_k)}{\|\mathbf{X}_j - \mathbf{X}_i\| \|\mathbf{X}_j - \mathbf{X}_k\|}\right) \end{aligned} \quad (69)$$

Its first and second derivative have already been derived in [40] and are recalled here for the sake of completeness. The associated Jacobian $\mathbf{G}$ is:

$$\begin{aligned} \mathbf{G} &= \begin{bmatrix} \frac{\partial \theta_{ijk}}{\partial \mathbf{V}_i} & \frac{\partial \theta_{ijk}}{\partial \mathbf{V}_j} & \frac{\partial \theta_{ijk}}{\partial \mathbf{V}_k} \end{bmatrix} \\ \frac{\partial \theta_{ijk}}{\partial \mathbf{V}_i} &= -\frac{1}{\sqrt{\gamma}} \tilde{\mathbf{r}}_i, \quad \frac{\partial \theta_{ijk}}{\partial \mathbf{V}_j} = \frac{1}{\sqrt{\gamma}} (\tilde{\mathbf{r}}_i + \tilde{\mathbf{r}}_k), \\ \frac{\partial \theta_{ijk}}{\partial \mathbf{V}_k} &= -\frac{1}{\sqrt{\gamma}} \tilde{\mathbf{r}}_k \end{aligned} \quad (70)$$

with the vectors

$$\begin{aligned} \tilde{\mathbf{r}}_i &= [\tilde{\mathbf{r}}_{jk} - \tilde{\mathbf{r}}_{ji}(\tilde{\mathbf{r}}_{ji}^T \tilde{\mathbf{r}}_{jk})]/\ell_{ji}, \quad \tilde{\mathbf{r}}_k = [(\tilde{\mathbf{r}}_{ji} - \tilde{\mathbf{r}}_{jk})(\tilde{\mathbf{r}}_{ji}^T \tilde{\mathbf{r}}_{jk})]/\ell_{jk}, \\ \tilde{\mathbf{r}}_{mn} &= \mathbf{x}_{mn}/\ell_{mn} \text{ and } \gamma = 1 - (\tilde{\mathbf{r}}_{ji} - \tilde{\mathbf{r}}_{jk})^2 \end{aligned}$$

The associated Hessian $\mathbf{H}$ is:

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_i \mathbf{V}_i} & \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_i \mathbf{V}_j} & \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_i \mathbf{V}_k} \\ \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_j \mathbf{V}_j} & \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_j \mathbf{V}_k} & \\ \text{sym.} & \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_k \mathbf{V}_k} & \end{bmatrix} \quad (71)$$

with the following derivatives:

$$\begin{aligned} \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_i \mathbf{V}_i} &= -\frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\sqrt{\gamma^3}} \bar{\mathbf{r}}_i \bar{\mathbf{r}}_i^T \\ &\quad + \frac{1}{\sqrt{\gamma} \ell_{ji}} \left(\bar{\mathbf{r}}_i \bar{\mathbf{r}}_{ji}^T + \bar{\mathbf{r}}_{ji} \bar{\mathbf{r}}_i^T + \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\ell_{ji}} \left(\mathbf{1} - \bar{\mathbf{r}}_{ji} \bar{\mathbf{r}}_{ji}^T \right) \right) \\ \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_i \mathbf{V}_j} &= \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\sqrt{\gamma^3}} \bar{\mathbf{r}}_i (\bar{\mathbf{r}}_i + \bar{\mathbf{r}}_k)^T \\ &\quad - \frac{1}{\sqrt{\gamma} \ell_{jk}} \left(\bar{\mathbf{r}}_i \bar{\mathbf{r}}_{ji}^T - \frac{1}{\ell_{jk}} \left(\mathbf{1} - \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_{jk}^T \right) \right. \\ &\quad \left. + \bar{\mathbf{r}}_{ji} (\bar{\mathbf{r}}_i + \bar{\mathbf{r}}_k)^T + \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\ell_{ji}} \left(\mathbf{1} - \bar{\mathbf{r}}_{ji} \bar{\mathbf{r}}_{ji}^T \right) \right) \\ \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_i \mathbf{V}_k} &= -\frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\sqrt{\gamma^3}} \bar{\mathbf{r}}_i \bar{\mathbf{r}}_k^T - \frac{1}{\sqrt{\gamma} \ell_{ji}} \left(\frac{1}{\ell_{jk}} \left(\mathbf{1} - \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_{jk}^T \right) - \bar{\mathbf{r}}_{ji} \bar{\mathbf{r}}_k^T \right) \\ \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_j \mathbf{V}_j} &= \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\sqrt{\gamma^3}} (\bar{\mathbf{r}}_i + \bar{\mathbf{r}}_k) (\bar{\mathbf{r}}_i + \bar{\mathbf{r}}_k)^T \\ &\quad + \frac{1}{\sqrt{\gamma} \ell_{ji}} \left(\bar{\mathbf{r}}_i \bar{\mathbf{r}}_{ji}^T + \frac{1}{\ell_{jk}} \left(\mathbf{1} - \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_{jk}^T \right) \right. \\ &\quad \left. + \bar{\mathbf{r}}_{ji} (\bar{\mathbf{r}}_i + \bar{\mathbf{r}}_k)^T + \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\ell_{ji}} \left(\mathbf{1} - \bar{\mathbf{r}}_{ji} \bar{\mathbf{r}}_{ji}^T \right) \right) \\ &\quad + \frac{1}{\sqrt{\gamma} \ell_{jk}} \left(\bar{\mathbf{r}}_k \bar{\mathbf{r}}_{jk}^T + \frac{1}{\ell_{ji}} \left(\mathbf{1} - \bar{\mathbf{r}}_{ji} \bar{\mathbf{r}}_{ji}^T \right) \right. \\ &\quad \left. + \bar{\mathbf{r}}_{jk} (\bar{\mathbf{r}}_k + \bar{\mathbf{r}}_i)^T + \frac{\bar{\mathbf{r}}_{jk}^T \bar{\mathbf{r}}_{ji}}{\ell_{jk}} \left(\mathbf{1} - \bar{\mathbf{r}}_{jl} \bar{\mathbf{r}}_{jl}^T \right) \right) \\ \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_j \mathbf{V}_k} &= \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\sqrt{\gamma^3}} (\bar{\mathbf{r}}_i + \bar{\mathbf{r}}_k) \bar{\mathbf{r}}_k^T \\ &\quad + \frac{1}{\sqrt{\gamma} \ell_{jk}} \left(\frac{1}{\ell_{ji}} \left(\mathbf{1} - \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_{jk}^T \right) - \bar{\mathbf{r}}_{ji} \bar{\mathbf{r}}_k^T \right) \\ &\quad - \frac{1}{\sqrt{\gamma} \ell_{jk}} \left(\bar{\mathbf{r}}_k \bar{\mathbf{r}}_{jk}^T + \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_k^T + \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\ell_{ji}} \left(\mathbf{1} - \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_{jk}^T \right) \right) \\ \frac{\partial^2 \theta_{ijk}}{\partial \mathbf{V}_k \mathbf{V}_k} &= -\frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\sqrt{\gamma^3}} \bar{\mathbf{r}}_k \bar{\mathbf{r}}_k^T \\ &\quad + \frac{1}{\sqrt{\gamma} \ell_{jk}} \left(\bar{\mathbf{r}}_i \bar{\mathbf{r}}_{jk}^T + \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_i^T + \frac{\bar{\mathbf{r}}_{ji}^T \bar{\mathbf{r}}_{jk}}{\ell_{ji}} \left(\mathbf{1} - \bar{\mathbf{r}}_{jk} \bar{\mathbf{r}}_{jk}^T \right) \right) \end{aligned} \quad (72)$$

Acknowledgements We gratefully thank PD Dr. Sigrun Ortleb for the fruitful discussion on the fundamentals of the mathematical derivation of the new master–slave elimination scheme.

Funding Open Access funding enabled and organized by Projekt DEAL.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Abel JF, Shephard MS (1979) An algorithm for multipoint constraints in finite element analysis. *Int J Numer Methods Eng* 14(3):464–467
2. Ainsworth M (2001) Essential boundary conditions and multi-point constraints in finite element analysis. *Comput Methods Appl Mech Eng* 190(48):6323–6339
3. Ansys Inc. (2020) Contact technology guide. Ansys documentation
4. Ansys Inc. (2020) Element reference. Ansys documentation
5. Areias P, Rabczuk T, Dias-da Costa D, Pires EB (2012) Implicit solutions with consistent additive and multiplicative components. *Finite Elem Anal Des* 57:15–31
6. Barlow J (1982) Constraint relationships in linear and nonlinear finite element analyses. *Int J Numer Methods Eng* 18(4):521–533
7. Bathe KJ, Dvorkin EN (1985) A four-node plate bending element based on Mindlin/Reissner plate theory and a mixed interpolation. *Int J Numer Methods Eng* 21(2):367–383
8. Belytschko T, Liu WK, Moran B, Elkhodary KI (2014) *Nonlinear finite elements for continua and structures*, 2nd edn. Wiley, Hoboken
9. Benzi M, Golub GH, Liesen J (2005) Numerical solution of saddle point problems. *Acta Numer* 14:1–137
10. Betsch P, Gruttmann F, Stein E (1996) A 4-node finite shell element for the implementation of general hyperelastic 3D-elasticity at finite strains. *Comput Methods Appl Mech Eng* 130(1):57–79
11. Bounghard J, Wackerfuß J. Identification, elimination and handling of redundant nonlinear multi-point constraints (in preparation)
12. Bourgat JF, Dumay JM, Glowinski R (1980) Large displacement calculations of flexible pipelines by finite element and nonlinear programming methods. *SIAM J Sci Stat Comput* 1(1):34–81
13. Chang SC, Lin TW (1988) Constraint relation implementation for finite element analysis from an element basis. *Adv Eng Softw* (1978) 10(4):191–194
14. Cheney EW, Kincaid D (2008) *Numerical mathematics and computing*, 6th edn. Thomson Brooks/Cole, Belmont
15. Curiskis JI, Valliappan S (1978) A solution algorithm for linear constraint equations in finite element analysis. *Comput Struct* 8(1):117–124
16. Dassault Systems (2023) Abaqus documentation
17. Dhondt G (2004) *The finite element method for three-dimensional thermomechanical applications*. Wiley, Hoboken

18. Felippa CA (1978) Iterative procedures for improving penalty function solutions of algebraic systems. *Int J Numer Methods Eng* 12(5):821–836
19. Felippa CA (2004) Introduction to finite element methods. Lecture notes. University of Colorado, Denver
20. Glowinski R, Lions JL, Trémolières R (1976) Analyse numérique des inéquations variationnelles. Méthodes mathématiques de l'informatique, vol 5. Dunod, Paris
21. Greene BE (1966) Application of generalized constraints in the stiffness method of structural analysis. *AIAA J* 4(9):1531–1537
22. Hestenes MR (1969) Multiplier and gradient methods. *J Optim Theory Appl* 4(5):303–320
23. Ho RJ, Meguid SA, Sauvé RG (2004) Novel coupling constraint technique for explicit finite element analysis. *Int J Comput Methods* 01(02):309–328
24. Hueck U, Schreyer HL (1992) The use of orthogonal projections to handle constraints with applications to incompressible four-node quadrilateral elements. *Int J Numer Methods Eng* 35(8):1633–1661
25. Intel (2022) Intel oneAPI math kernel library. Developer reference
26. Irons BM, Draper KJ (1965) Lagrange multiplier techniques in structural analysis. *AIAA J* 3(6):1172–1174
27. Jelenić G, Crisfield MA (1996) Non-linear 'master-slave' relationships for joints in 3-d beams with large rotations. *Comput Methods Appl Mech Eng* 135(3–4):211–228
28. Kruse C, Darrigrand V, Tardieu C, Arioli M, Rüde U (2020) Application of an iterative Golub–Kahan algorithm to structural mechanics problems with multi-point constraints. *Adv Model Simul Eng Sci* 7(1):45
29. Lagrange JL (1815) Mécanique analytique
30. Muñoz JJ (2008) On the modelling of incompressibility in linear and non-linear elasticity with the master–slave approach. *Int J Numer Methods Eng* 74(2):269–293
31. Narayanaswamy OS (1985) Processing nonlinear multipoint constraints in the finite element method. *Int J Numer Methods Eng* 21(7):1283–1288
32. Polizzi E (2009) Density-matrix-based algorithm for solving eigenvalue problems. *Phys Rev B* 79:115112
33. Powell MJD (1969) A method for nonlinear constraints in minimization problems. University Keele, Keele
34. Saint-Georges P, Notay Y, Warzée G (1998) Efficient iterative solution of constrained finite element analyses. *Comput Methods Appl Mech Eng* 160(1–2):101–114
35. Schreyer HL, Parsons DA (1995) Direct application of constraints to symmetric algebraic systems. *Commun Numer Methods Eng* 11(7):563–573
36. Shephard MS (1984) Linear multipoint constraints applied via transformation as part of a direct stiffness assembly process. *Int J Numer Methods Eng* 20(11):2107–2112
37. Simo JC, Wriggers P, Taylor RL (1985) A perturbed Lagrangian formulation for the finite element solution of contact problems. *Comput Methods Appl Mech Eng* 50(2):163–180
38. Strassen V (1969) Gaussian elimination is not optimal. *Numer Math* 13(4):354–356
39. Szabo BA, Tsai CT (1973) The quadratic programming approach to the finite element method. *Int J Numer Methods Eng* 5(3):375–381
40. Wackerfuß J (2009) Molecular mechanics in the context of the finite element method. *Int J Numer Methods Eng* 77(7):969–997
41. Wagner W, Gruttmann F (2005) A robust non-linear mixed hybrid quadrilateral shell element. *Int J Numer Methods Eng* 64(5):635–666
42. Webb JP (1990) Imposing linear constraints in finite-element analysis. *Commun Appl Numer Methods* 6(6):471–475
43. Wojtyra M, Frączek J (2013) Comparison of selected methods of handling redundant constraints in multibody systems simulations. *J. Comput. Nonlinear Dyn.* 8(2):021007
44. Xikui L, Guoqiang L (1988) A new approach for finite element analysis of constrained problems in elasticity. *Comput. Struct.* 30(4):915–918
45. Zienkiewicz OC (1974) Constrained variational principles and penalty function methods in finite element analysis. In: Watson GA (ed) Conference on the numerical solution of differential equations, vol 363. Lecture notes in mathematics. Springer, Berlin, pp 207–214

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.